Riemanni hüpoteesist

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Part I
hüpotees

$\exists_{x_1}($
$\zeta(x_1)=0 \wedge$
$x_1 \in C \wedge$
$\neg(Re(x_1)=1/2) \wedge$
$\neg(\exists_{x_2}(x_2 \in N \wedge Re(x_1)=-x_2*2 \wedge Im(x_1)=0)$
$)$

Part II
muud seoud väited

1 zetafunktsioonist

1.1 kui reaalosa suurem kui 1

$\forall_{x_1}(x_1 \in C \wedge Re(x_1)<1 \rightarrow \zeta(-x_1)=\sum_{n=1}^{\infty}(n^{x_1})=\sum_{n=1}^{\infty}(n^{x_1})=\sum_{n=1}^{\infty}(n^{Re(x_1)+i*Im(x_1)})=\sum_{n=1}^{\infty}(n^{Re(x_1)}*n^{i*Im(x_1)})=$
$\sum_{n=1}^{\infty}(n^{Re(x_1)}*(cos(Im(x_1)*ln(n))+i*sin(Im(x_1)*ln(n)))))$

1.2 Dirichlet series

avaldis1: Dirichlet_series_Wikipediast

$\forall_{x_1}(x_1 \in C \wedge Re(x_1)>0 \rightarrow \zeta(x_1)=(x_1-1)^{-1}*\sum_{n=1}^{\infty}(\frac{n}{(n+1)^{x_1}}-\frac{n-x_1}{n^{x_1}})$

lihtsustan seda avaldist:

$\forall_{x_1}(\forall_{x_2}(x_1 \in R \wedge x_2 \in R \wedge x_1>0 \rightarrow$
$\zeta(x_1+i*x_2)=$

$(x_1-1+i*x_2)^{-1}*\sum_{n=1}^{\infty}(\frac{n}{(n+1)^{x_1+i*x_2}}-\frac{n-x_1-i*x_2}{n^{x_1+i*x_2}})=$
$(x_1-1+i*x_2)^{-1}*\sum_{n=1}^{\infty}(\frac{n}{(n+1)^{x_1}*(cos(x_2*ln(n+1))+i*sin(x_2*ln(n+1)))}-\frac{n-x_1-i*x_2}{n^{x_1}*(cos(x_2*ln(n))+i*sin(x_2*ln(n)))})=$

$\frac{x_1-1-i*x_2}{(x_1-1)^2+x_2^2}*\sum_{n=1}^{\infty}(n*(n+1)^{-x_1}*(cos(x_2*ln(n+1))-i*sin(x_2*ln(n+1)))-(n-x_1-i*x_2)*n^{-x_1}*(cos(x_2*$
$ln(n))-i*sin(x_2*ln(n))))=$

$\sum_{n=1}^{\infty}(((x_1-1)-i*x_2)*(cos(x_2*ln(n+1))-i*sin(x_2*ln(n+1)))*((x_1-1)^2+x_2^2)^{-1}*n*(n+1)^{-x_1}-$
$((x_1-1)-i*x_2)*(cos(x_2*ln(n))-i*sin(x_2*ln(n)))*((x_1-1)^2+x_2^2)^{-1}*(n-x_1-i*x_2)*n^{-x_1})=$

$\sum_{n=1}^{\infty}((((x_1-1)*cos(x_2*ln(n+1))-x_2*sin(x_2*ln(n+1)))-i*((x_1-1)*sin(x_2*ln(n+1))+x_2*cos(x_2*ln(n+1))))*((x_1-1)^2+x_2^2)^{-1}*n*(n+1)^{-x_1}-$
$((n-x_1)*((x_1-1)*cos(x_2*ln(n))-x_2*sin(x_2*ln(n)))-x_2*((x_1-1)*sin(x_2*ln(n))+x_2*cos(x_2*ln(n)))-i*$
$((n-x_1)*((x_1-1)*sin(x_2*ln(n))+x_2*cos(x_2*ln(n)))+x_2*(((x_1-1)*cos(x_2*ln(n))-x_2*sin(x_2*ln(n))))))*$
$((x_1-1)^2+x_2^2)^{-1}*n^{-x_1})=$

$\sum_{n=1}^{\infty}(((x_1-1)*cos(x_2*ln(n+1))-x_2*sin(x_2*ln(n+1)))*((x_1-1)^2+x_2^2)^{-1}*n*(n+1)^{-x_1}-((n-x_1)*((x_1-1)*cos(x_2*ln(n))-x_2*sin(x_2*ln(n)))-x_2*((x_1-1)*sin(x_2*ln(n))+x_2*cos(x_2*ln(n))))*((x_1-1)^2+x_2^2)^{-1}*n^{-x_1}$
$-i*(((x_1-1)*sin(x_2*ln(n+1))+x_2*cos(x_2*ln(n+1)))*((x_1-1)^2+x_2^2)^{-1}*n*(n+1)^{-x_1}+((n-x_1)*((x_1-1)*sin(x_2*ln(n))+x_2*cos(x_2*ln(n)))+x_2*(((x_1-1)*cos(x_2*ln(n))-x_2*sin(x_2*ln(n)))))*((x_1-1)^2+x_2^2)^{-1}*n^{-x_1}))=$
$((x_1-1)^2+x_2^2)^{-1}*\sum_{n=1}^{\infty}(((x_1-1)*cos(x_2*ln(n+1))-x_2*sin(x_2*ln(n+1)))*n*(n+1)^{-x_1}-((n-x_1)*((x_1-1)*cos(x_2*ln(n))-x_2*sin(x_2*ln(n)))-x_2*((x_1-1)*sin(x_2*ln(n))+x_2*cos(x_2*ln(n))))*n^{-x_1})-i*((x_1-1)^2+x_2^2)^{-1}*\sum_{n=1}^{\infty}((((x_1-1)*sin(x_2*ln(n+1))+x_2*cos(x_2*ln(n+1)))*n*(n+1)^{-x_1}-((n-x_1)*((x_1-1)*sin(x_2*ln(n))+x_2*cos(x_2*ln(n)))+x_2*((x_1-1)*cos(x_2*ln(n))-x_2*sin(x_2*ln(n))))*n^{-x_1}))=$
$))$

järgneva väite nimi on zeta2_reaalosa_ja_imaginaarosa_eraldatud:

$\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\rightarrow$
$\zeta(x_1+i*x_2)=$
$((x_1-1)^2+x_2^2)^{-1}*\sum_{n=1}^{\infty}($
$((x_1-1)*cos(x_2*ln(n+1))-x_2*sin(x_2*ln(n+1)))*n*(n+1)^{-x_1}-$
$((n-x_1)*((x_1-1)*cos(x_2*ln(n))-x_2*sin(x_2*ln(n)))-x_2*((x_1-1)*sin(x_2*ln(n))+x_2*cos(x_2*ln(n))))*n^{-x_1}$
$)$
$-i*((x_1-1)^2+x_2^2)^{-1}*\sum_{n=1}^{\infty}($
$((x_1-1)*sin(x_2*ln(n+1))+x_2*cos(x_2*ln(n+1)))*n*(n+1)^{-x_1}-$
$((n-x_1)*((x_1-1)*sin(x_2*ln(n))+x_2*cos(x_2*ln(n)))+x_2*((x_1-1)*cos(x_2*ln(n))-x_2*sin(x_2*ln(n))))*n^{-x_1}$
$)=$
$))$

lihtsustan seda väidet edasi

$\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\rightarrow$
$\zeta(x_1+i*x_2)=$

$(x_1^2-2*x_1+1+x_2^2)^{-1}*\sum_{n=1}^{\infty}($
$sin(x_2*ln(n)+0*(2\pi)/4)*x_2*(n-1)*n^{-x_1}-sin(x_2*ln(n+1)+0*(2\pi)/4)*x_2*n*(n+1)^{-x_1}$
$+sin(x_2*ln(n)+1*(2\pi)/4)*(x_1^2-x_1-n*x_1+n+x_2^2)*n^{-x_1}$
$-sin(x_2*ln(n+1)+1*(2\pi)/4)*(1-x_1)*n*(n+1)^{-x_1}$
$)$
$+i*(x_1^2-2*x_1+1+x_2^2)^{-1}*\sum_{n=1}^{\infty}($
$sin(x_2*ln(n)+1*(2\pi)/4)*x_2*(n-1)*n^{-x_1}-sin(x_2*ln(n+1)+1*(2\pi)/4)*x_2*n*(n+1)^{-x_1}$
$+sin(x_2*ln(n)+2*(2\pi)/4)*(x_1^2-x_1-n*x_1+n+x_2^2)*n^{-x_1}$
$-sin(x_2*ln(n+1)+2*(2\pi)/4)*(1-x_1)*n*(n+1)^{-x_1}$
$)$
$=$

$\lim_{M\rightarrow\infty}(\sum_{n=1}^{M}($
$+sin(x_2*ln(n)+1*(2\pi)/4)*(x_1^2-x_1-n*x_1+n+x_2^2)*n^{-x_1}$
$-sin(x_2*ln(n+1)+1*(2\pi)/4)*(1-x_1)*n*(n+1)^{-x_1}$
$)-sin(x_2*ln(M+1)+0*(2\pi)/4)*x_2*M*(M+1)^{-x_1})*$
$(x_1^2-2*x_1+1+x_2^2)^{-1}$
$+i*\lim_{M\rightarrow\infty}(\sum_{n=1}^{M}($
$sin(x_2*ln(n)+2*(2\pi)/4)*(x_1^2-x_1-n*x_1+n+x_2^2)*n^{-x_1}$
$-sin(x_2*ln(n+1)+2*(2\pi)/4)*(1-x_1)*n*(n+1)^{-x_1}$
$)-sin(x_2*ln(M+1)+1*(2\pi)/4)*x_2*M*(M+1)^{-x_1})*$
$(x_1^2-2*x_1+1+x_2^2)^{-1}=$

$\lim_{M\to\infty}(\sum_{n=1}^{M}($
$+sin(x_2*ln(n)+1*(2\pi)/4)*(x_1^2-2*x_1+x_2^2+1)*n^{-x_1}$
$+sin(x_2*ln(n)+1*(2\pi)/4)*(1-x_1)*(n-1)*n^{-x_1}-sin(x_2*ln(n+1)+1*(2\pi)/4)*(1-x_1)*n*(n+1)^{-x_1}$
$)-sin(x_2*ln(M+1)+0*(2\pi)/4)*x_2*M*(M+1)^{-x_1})*$
$(x_1^2-2*x_1+1+x_2^2)^{-1}$
$+i*\lim_{M\to\infty}(\sum_{n=1}^{M}($
$+sin(x_2*ln(n)+2*(2\pi)/4)*(x_1^2-2*x_1+x_2^2+1)*n^{-x_1}$
$+sin(x_2*ln(n)+2*(2\pi)/4)*(1-x_1)*(n-1)*n^{-x_1}-sin(x_2*ln(n+1)+2*(2\pi)/4)*(1-x_1)*n*(n+1)^{-x_1}$
$)-sin(x_2*ln(M+1)+1*(2\pi)/4)*x_2*M*(M+1)^{-x_1})*$
$(x_1^2-2*x_1+1+x_2^2)^{-1}=$
$\lim_{M\to\infty}($
$\sum_{n=1}^{M}(sin(x_2*ln(n)+1*(2\pi)/4)*n^{-x_1})$
$-(sin(x_2*ln(M+1)+1*(2\pi)/4)*(1-x_1)+sin(x_2*ln(M+1)+0*(2\pi)/4)*x_2)*M*(M+1)^{-x_1}*((1-x_1)^2+$
$x_2^2)^{-1})$

$+i*\lim_{M\to\infty}($
$\sum_{n=1}^{M}(sin(x_2*ln(n)+2*(2\pi)/4)*n^{-x_1})$
$-(sin(x_2*ln(M+1)+2*(2\pi)/4)*(1-x_1)+sin(x_2*ln(M+1)+1*(2\pi)/4)*x_2)*M*(M+1)^{-x_1}*((1-x_1)^2+$
$x_2^2)^{-1})$
$=$

$\lim_{M\to\infty}($
$\sum_{n=1}^{M}(sin(x_2*ln(n)+1*(2\pi)/4)*n^{-x_1})$
$-(sin(x_2*ln(M+1)+1*(2\pi)/4)*(1-x_1)+sin(x_2*ln(M+1)+0*(2\pi)/4)*x_2)*M*(M+1)^{-x_1}*(1+1/M)*$
$((1-x_1)^2+x_2^2)^{-1})$

$+i*\lim_{M\to\infty}($
$\sum_{n=1}^{M}(sin(x_2*ln(n)+2*(2\pi)/4)*n^{-x_1})$
$-(sin(x_2*ln(M+1)+2*(2\pi)/4)*(1-x_1)+sin(x_2*ln(M+1)+1*(2\pi)/4)*x_2)*M*(M+1)^{-x_1}*(1+1/M)*$
$((1-x_1)^2+x_2^2)^{-1})$
$=$

$\lim_{M\to\infty}($
$\sum_{n=1}^{M}(sin(x_2*ln(n)+1*(2\pi)/4)*n^{-x_1})$
$-(sin(x_2*ln(M+1)+1*(2\pi)/4)*(1-x_1)+sin(x_2*ln(M+1)+0*(2\pi)/4)*x_2)*M^{1-x_1}*((1-x_1)^2+x_2^2)^{-1}$
$)$
$+i*\lim_{M\to\infty}($
$\sum_{n=1}^{M}(sin(x_2*ln(n)+2*(2\pi)/4)*n^{-x_1})$
$-(sin(x_2*ln(M+1)+2*(2\pi)/4)*(1-x_1)+sin(x_2*ln(M+1)+1*(2\pi)/4)*x_2)*M^{1-x_1}*((1-x_1)^2+x_2^2)^{-1}$
$)=$

))

Valem zeta9_M_plus_1e_ei_ole:

$\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\to$
$\zeta(x_1+i*x_2)=$

$\lim_{M\to\infty}($
$\sum_{n=1}^{M}(sin(x_2*ln(n)+1*(2\pi)/4)*n^{-x_1})$
$-(sin(x_2*ln(M)+1*(2\pi)/4)*(1-x_1)+sin(x_2*ln(M)+0*(2\pi)/4)*x_2)*M^{1-x_1}*((1-x_1)^2+x_2^2)^{-1}$
$)$
$+i*\lim_{M\to\infty}($
$\sum_{n=1}^{M}(sin(x_2*ln(n)+2*(2\pi)/4)*n^{-x_1})$
$-(sin(x_2*ln(M)+2*(2\pi)/4)*(1-x_1)+sin(x_2*ln(M)+1*(2\pi)/4)*x_2)*M^{1-x_1}*((1-x_1)^2+x_2^2)^{-1}$
$)$

))

kuivõrd M läheneb lõpmatusele võin igal pool avaldises i asemele panna $floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})$ või floor(e$^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi/x_2}$).

$\forall_{q_1}(\forall_{q_2}(\forall_{x_1}(\forall_{x_2}(x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \rightarrow$
$\zeta(x_1 + i*x_2) =$

$\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})}(sin(x_2*ln(n)+1*(2\pi)/4)*n^{-x_1})$
$-(sin(x_2*ln(floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi}))+1*(2\pi)/4)*(1-x_1)+sin(x_2*ln(floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi}))+$
$0*(2\pi)/4)*x_2)*floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})^{1-x_1}*((1-x_1)^2+x_2^2)^{-1}$
$)$
$+i*\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})}(sin(x_2*ln(n)+2*(2\pi)/4)*n^{-x_1})$
$-(sin(x_2*ln(floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi}))+2*(2\pi)/4)*(1-x_1)+sin(x_2*ln(floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi}))+$
$1*(2\pi)/4)*x_2)*floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})^{1-x_1}*((1-x_1)^2+x_2^2)^{-1}$
$) =$

$\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})}(sin(x_2*ln(n)+1*(2\pi)/4)*n^{-x_1})$
$-(sin(x_2*ln((e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi}))+1*(2\pi)/4)*(1-x_1)+sin(x_2*ln((e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi}))+0*(2\pi)/4)*$
$x_2)*(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})^{1-x_1}*((1-x_1)^2+x_2^2)^{-1}$
$)$
$+i*\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})}(sin(x_2*ln(n)+2*(2\pi)/4)*n^{-x_1})$
$-(sin(x_2*ln((e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi}))+2*(2\pi)/4)*(1-x_1)+sin(x_2*ln((e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi}))+1*(2\pi)/4)*$
$x_2)*(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})^{1-x_1}*((1-x_1)^2+x_2^2)^{-1}$
$) =$

$\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})}(sin(x_2*ln(n)+1*(2\pi)/4)*n^{-x_1})$
$-(sin((floor(M)*(x_2/abs(x_2))+q_1)*2\pi+1*(2\pi)/4)*(1-x_1)+sin((floor(M)(x_2/abs(x_2))+q_1)*2\pi+0*$
$(2\pi)/4)*x_2)*(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi*(1-x_1)})*((1-x_1)^2+x_2^2)^{-1}$
$)$
$+i*\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})}(sin(x_2*ln(n)+2*(2\pi)/4)*n^{-x_1})$
$-(sin((floor(M)*(x_2/abs(x_2))+q_2)*2\pi+2*(2\pi)/4)*(1-x_1)+sin((floor(M)*(x_2/abs(x_2))+q_2)*2\pi+1*$
$(2\pi)/4)*x_2)*(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi*(1-x_1)})*((1-x_1)^2+x_2^2)^{-1}$
$) =$
))))

zeta_funktsioon_2_q_ga:

$\forall_{q_1}(\forall_{q_2}(\forall_{x_1}(\forall_{x_2}(x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \rightarrow$
$\zeta(x_1 + i*x_2) =$
$\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})}(sin(x_2*ln(n)+(2\pi)*1/4)*n^{-x_1})$
$-(sin((q_1+1/4)*2\pi)*(1-x_1)+sin((q_1+0/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}$
$)$
$+i*\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})}(sin(x_2*ln(n)+(2\pi)*2/4)*n^{-x_1})$
$-(sin((q_2+2/4)*2\pi)*(1-x_1)+sin((q_2+1/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}$
$) =$

))))

Nagu kvantrite abilgi öeldud on kehtib võrrand mistahes q_1 ja q_2 väärtuste puhul, aga kuna pole kindel, et millised väärtused q_1 ja q_2 asemele pannes saab kõige paremal kujul avaldise, siis loon su subsectionites võrrandist erinevaid variante, kus q_1 ja q_2'el on erinevad väärtused.

1.2.1 integraliga

kasutades seost $a*cos(t)+b*sin(t)=\sqrt{a^2+b^2}*sin(t+\arctan(\frac{a}{b}))$ lihtsustan eelnevat valemit:

$\forall_{q_1}(\forall_{q_2}(\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\rightarrow$
$\zeta(x_1+i*x_2)=$
$\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})}(sin(x_2*ln(n)+(2\pi)*1/4)*n^{-x_1})$
$-sin(q_1*2\pi+\arctan((1-x_1)/x_2)*e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1/2}$
$)$
$+i*\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})}(sin(x_2*ln(n)+(2\pi)*2/4)*n^{-x_1})$
$-sin(q_2*2\pi-\arctan(x_2/(1-x_1)))*e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1/2}$
$)$
))))

igalt summa elemendilt lahutan integraali eelmisest elemendist selle elemendine ja liidan maha lahutatud osa peale summerimist uuesti tagasi. Mingil põhjusel, mille chatgpt kirjutas tomib see ainult tingimusel $x_1<1$.

$\forall_{q_1}(\forall_{q_2}(\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\wedge x_1<1\rightarrow$
$\zeta(x_1+i*x_2)=\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})}(sin(x_2*ln(n)+(2\pi)*1/4)*n^{-x_1}-\int_{n-1}^{n}(dy*sin(x_2*ln(y)+(2\pi)*1/4)*y^{-x_1}))$
$+\int_0^{e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi}}(dy*sin(x_2*ln(y)+(2\pi)*1/4)*y^{-x_1})$
$+\int_{e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi}}^{floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi}-1)}(dy*sin(x_2*ln(y)+(2\pi)*1/4)*y^{-x_1})$
$-sin(q_1*2\pi+\arctan((1-x_1)/x_2)*e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1/2}$
$)$
$+i*\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})}(sin(x_2*ln(n)+(2\pi)*2/4)*n^{-x_1}-\int_{n-1}^{n}(dy*sin(x_2*ln(y)+(2\pi)*2/4)*y^{-x_1}))$
$+\int_0^{e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi}}(dy*sin(x_2*ln(y)+(2\pi)*2/4)*y^{-x_1})$
$+\int_{e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi}}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})-1}(dy*sin(x_2*ln(y)+(2\pi)*2/4)*y^{-x_1})$
$-sin(q_2*2\pi-\arctan(x_2/(1-x_1)))*e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1/2}$
$)$
))))

asendan integrali sellega võrdse arihmeetilise avaldisega.

$\forall_{q_1}(\forall_{q_2}(\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\wedge x_1<1\rightarrow$
$\zeta(x_1+i*x_2)=\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})}($
$sin(x_2*ln(n)+(2\pi)*1/4)*n^{-x_1}$
$-(n^{1-x_1}*sin(x_2*ln(n)+arctan((1-x_1)/x_2))-(n-1)^{1-x_1}*sin(x_2*ln(n-1)+arctan((1-x_1)/x_2)))*((1-x_1)^2+x_2^2)^{-1/2}$
$)$
$+((e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})^{1-x_1}*sin(x_2*ln(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})+arctan((1-x_1)/x_2))-0^{1-x_1}*sin(x_2*ln(0)+arctan((1-x_1)/x_2)))*((1-x_1)^2+x_2^2)^{-1/2}$

$+((floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi}-1))^{1-x_1}*sin(x_2*ln(floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi}-1))+arctan((1-x_1)/x_2))-(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})^{1-x_1}*sin(x_2*ln(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})+arctan((1-x_1)/x_2)))*((1-x_1)^2+x_2^2)^{-1/2}$

$-sin(q_1*2\pi+\arctan((1-x_1)/x_2)*e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1/2}$
)
$+i*\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})}($
$sin(x_2*ln(n)+(2\pi)*2/4)*n^{-x_1}-$
$(n^{1-x_1}*sin(x_2*ln(n)-arctan(x_2/(1-x_1)))+(n-1)^{1-x_1}*sin(x_2*ln(n-1)-arctan(x_2/(1-x_1))))*((1-x_1)^2+x_2^2)^{-1/2}$
)
$+((e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})^{1-x_1}*sin(x_2*ln(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})-arctan(x_2/(1-x_1)))+0^{1-x_1}*sin(x_2*ln(0)-arctan(x_2/(1-x_1))))*((1-x_1)^2+x_2^2)^{-1/2}$

$+((floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})-1)^{1-x_1}*sin(x_2*ln(floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})-1)-arctan(x_2/(1-x_1)))+(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})^{1-x_1}*sin(x_2*ln(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})-arctan(x_2/(1-x_1))))*((1-x_1)^2+x_2^2)^{-1/2}$
$-sin(q_2*2\pi-\arctan(x_2/(1-x_1)))*e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1/2}$
)
))))

lihtsustan. Kuna kui $x_1<1$ siis $\lim_{y\to0^+}(y^{1-x_1}*sin(x_2*ln(y)+arctan((1-x_1)/x_2)))=0$ ($0^{1-x_1}*sin(x_2*ln(0)+arctan((1-x_1)/x_2))=0$), sest $sin(x_2*ln(0)+arctan((1-x_1)/x_2))$ on määramatu, aga kindlalt piirides -1 kuni 1.

Kuna kui $x_1<1$ siis $\lim_{y\to0^+}(y^{1-x_1}*sin(x_2*ln(y)+arctan(x_2/(1-x_1))))=0$ ($0^{1-x_1}*sin(x_2*ln(0)-arctan(x_2/(1-x_1)))=0$),sest $sin(x_2*ln(0)-arctan(x_2/(1-x_1)))$ on määrmatu, aga kindlalt piirides -1 kuni 1.

SIIN OLI VIGA! kas nüüd korras?

$\forall_{q_1}(\forall_{q_2}(\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\wedge x_1<1\to$
$\zeta(x_1+i*x_2)=\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})}($
$sin(x_2*ln(n)+(2\pi)*1/4)*n^{-x_1}$
$-sin(x_2*ln(n)+arctan((1-x_1)/x_2))*n^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}$
$+sin(x_2*ln(n-1)+arctan((1-x_1)/x_2))*(n-1)^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}$
)
$+sin(q_1*2\pi+arctan((1-x_1)/x_2))*e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1/2}$

$+((floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi}-1))^{1-x_1}*sin(x_2*ln(floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi}-1))+arctan((1-x_1)/x_2))-(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})^{1-x_1}*sin(x_2*ln(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})+arctan((1-x_1)/x_2)))*((1-x_1)^2+x_2^2)^{-1/2}$

$-sin(q_1*2\pi+\arctan((1-x_1)/x_2)*e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1/2}$
)
$+i*\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})}($
$sin(x_2*ln(n)+(2\pi)*2/4)*n^{-x_1}$
$-sin(x_2*ln(n)-arctan(x_2/(1-x_1)))*n^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}$
$+sin(x_2*ln(n-1)-arctan(x_2/(1-x_1)))*(n-1)^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}$
)
$+sin(q_2*2\pi-arctan(x_2/(1-x_1)))*e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1/2}$

$+((floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})-1)^{1-x_1}*sin(x_2*ln(floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})-1)-arctan(x_2/(1-x_1)))+(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})^{1-x_1}*sin(q_2*2\pi-arctan(x_2/(1-x_1))))*((1-x_1)^2+x_2^2)^{-1/2}$
$-sin(q_2*2\pi-\arctan(x_2/(1-x_1)))*e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1/2}$
)
))))

taandan

$\forall_{q_1}(\forall_{q_2}(\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\wedge x_1<1\to$
$\zeta(x_1+i*x_2)=\lim_{M\to\infty}($

$\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})}($
$sin(x_2*ln(n)+(2\pi)*1/4)*n^{-x_1}$
$-sin(x_2*ln(n)+arctan((1-x_1)/x_2))*n^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}$
$+sin(x_2*ln(n-1)+arctan((1-x_1)/x_2))*(n-1)^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}$
$)$
$+($
$sin(x_2*ln(floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi}-1))+arctan((1-x_1)/x_2))*(floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi}-1))^{1-x_1}$
$-sin(x_2*ln(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})+arctan((1-x_1)/x_2))*(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})^{1-x_1}$
$)*((1-x_1)^2+x_2^2)^{-1/2}$

$)$
$+i*\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})}($
$sin(x_2*ln(n)+(2\pi)*2/4)*n^{-x_1}$
$-sin(x_2*ln(n)-arctan(x_2/(1-x_1)))*n^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}$
$+sin(x_2*ln(n-1)-arctan(x_2/(1-x_1)))*(n-1)^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}$
$)$
$+($
$sin(x_2*ln(floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})-1)-arctan(x_2/(1-x_1)))*(floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})-1)^{1-x_1}$
$+sin(q_2*2\pi-arctan(x_2/(1-x_1)))*(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})^{1-x_1}$
$)*((1-x_1)^2+x_2^2)^{-1/2}$
$)$
$))))$

kuna kui M on väga suur, siis $\lim_{M\to 0}(ln(M-2)-ln(M)=0)\ \wedge\ \lim_{M\to 0}(M^x-(M-2)^x)=0$, siis

$sin(x_2*ln(floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi}-1))+arctan((1-x_1)/x_2))*(floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi}-1))^{1-x_1}$
$-sin(x_2*ln(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})+arctan((1-x_1)/x_2))*(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})^{1-x_1}=0$

ja

$sin(x_2*ln(floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})-1)-arctan(x_2/(1-x_1)))*(floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})-1)^{1-x_1}$
$+sin(q_2*2\pi-arctan(x_2/(1-x_1)))*(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})^{1-x_1}=0$

seega:

$\forall_{q_1}(\forall_{q_2}(\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\wedge x_1<1\to\zeta(x_1+i*x_2)=$
$\lim_{M\to\infty}(\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})}($
$sin(x_2*ln(n)+(2\pi)*1/4)*n^{-x_1}$
$-sin(x_2*ln(n)+arctan((1-x_1)/x_2))*n^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}$
$+sin(x_2*ln(n-1)+arctan((1-x_1)/x_2))*(n-1)^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}$
$))$
$+i*\lim_{M\to\infty}(\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})}($
$sin(x_2*ln(n)+(2\pi)*2/4)*n^{-x_1}$
$-sin(x_2*ln(n)-arctan(x_2/(1-x_1)))*n^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}$
$+sin(x_2*ln(n-1)-arctan(x_2/(1-x_1)))*(n-1)^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}$
$))$
$))))$

kuna vastavate q_1 ja q_2 väärtuste valimise korral saab summa piiri suvaliselt nihutada, siis võib summa piiriks lihtsalt Mi panna

$\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\wedge x_1<1\to\zeta(x_1+i*x_2)=$
$\lim_{M\to\infty}(\sum_{n=1}^{M}($
$sin(x_2*ln(n)+(2\pi)*1/4)*n^{-x_1}$
$-sin(x_2*ln(n)+arctan((1-x_1)/x_2))*n^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}$
$+sin(x_2*ln(n-1)+arctan((1-x_1)/x_2))*(n-1)^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}$
$))$
$+i*\lim_{M\to\infty}(\sum_{n=1}^{M}($
$-sin(x_2*ln(n))*n^{-x_1}$

$$
-sin(x_2*ln(n)-arctan(x_2/(1-x_1)))*n^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}
$$
$$
+sin(x_2*ln(n-1)-arctan(x_2/(1-x_1)))*(n-1)^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}
$$
$$
))
$$
$$
))
$$

Muudan summa piiri 1est 0iks:

$$
\forall_{x_1}(\forall_{x_2}(x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \wedge x_1 < 1 \rightarrow \zeta(x_1+i*x_2) =
$$
$$
\lim_{M\to\infty}(\sum_{n=0}^{M}(
$$
$$
sin(x_2*ln(n+1)+(2\pi)*1/4)*(n+1)^{-x_1}
$$
$$
-sin(x_2*ln(n+1)+arctan((1-x_1)/x_2))*(n+1)^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}
$$
$$
+sin(x_2*ln(n)+arctan((1-x_1)/x_2))*n^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}
$$
$$
))
$$
$$
+i*\lim_{M\to\infty}(\sum_{n=0}^{M}(
$$
$$
-sin(x_2*ln(n+1))*(n+1)^{-x_1}
$$
$$
-sin(x_2*ln(n+1)-arctan(x_2/(1-x_1)))*(n+1)^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}
$$
$$
+sin(x_2*ln(n)-arctan(x_2/(1-x_1)))*n^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}
$$
$$
))
$$
$$
))
$$

lihtsustan

$$
\forall_{x_1}(\forall_{x_2}(x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \wedge x_1 < 1 \rightarrow \zeta(x_1+i*x_2) =
$$
$$
\lim_{M\to\infty}(\sum_{n=0}^{M}(
$$
$$
sin(x_2*ln(n+1)+(2\pi)*1/4)*(n+1)^{-x_1}
$$
$$
-sin(x_2*ln(n+1)+arctan((1-x_1)/x_2))*(n+1)^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}
$$
$$
+sin(x_2*ln(n)+arctan((1-x_1)/x_2))*n^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}
$$
$$
))
$$
$$
-i*\lim_{M\to\infty}(\sum_{n=0}^{M}(
$$
$$
sin(x_2*ln(n+1))*(n+1)^{-x_1}
$$
$$
-sin(x_2*ln(n+1)+arctan((1-x_1)/x_2))*(n+1)^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}
$$
$$
+sin(x_2*ln(n)+arctan((1-x_1)/x_2))*n^{1-x_1}*((1-x_1)^2+x_2^2)^{-1/2}
$$
$$
))
$$
$$
))
$$

1.2.2 edasine lihtsustamine lihtsamaid funktsioone sisaldama.

asendan siinuse selle Taylori reaga:

$$
\forall_{q_1}(\forall_{q_2}(\forall_{x_1}(\forall_{x_2}(x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \rightarrow
$$
$$
\zeta(x_1+i*x_2) =
$$
$$
\lim_{M\to\infty}(
$$
$$
\sum_{n_0=1}^{floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})}(
$$
$$
{n_0}^{-x_1}*\sum_{n_1=0}^{\infty}((-1)^{n_1}*(x_2*ln(n_0))^{2*n_1}*factorial(2*n_1)^{-1})
$$
$$
)
$$
$$
-(sin((q_1+1/4)*2\pi)*(1-x_1)+sin((q_1+0/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}
$$
$$
)
$$
$$
+i*\lim_{M\to\infty}(
$$
$$
-\sum_{n_0=1}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})}(
$$
$$
{n_0}^{-x_1}*\sum_{n_1=0}^{\infty}((-1)^{n_1+}*(x_2*ln(n_0))^{2*n_1+1}*factorial(2*n_1+1)^{-1})
$$
$$
)
$$
$$
-(sin((q_2+2/4)*2\pi)*(1-x_1)+sin((q_2+1/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}
$$
$$
)
$$
$$
))))
$$

muudan tegurite järjekorda:

$$
\forall_{q_1}(\forall_{q_2}(\forall_{x_1}(\forall_{x_2}(x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \rightarrow
$$
$$
\zeta(x_1+i*x_2) =
$$
$$
\lim_{M\to\infty}(
$$

$\sum_{n_0=1}^{floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})}(n_0{}^{-x_1}*$
$\sum_{n_1=0}^{\infty}((-1)^{n_1}*x_2^{2*n_1}*factorial(2*n_1)^{-1}*ln(n_0)^{2*n_1})$
$)$
$-(sin((q_1+1/4)*2\pi)*(1-x_1)+sin((q_1+0/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}$
$)$
$+i*\lim_{M\to\infty}($
$-\sum_{n_0=1}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})}(n_0{}^{-x_1}*$
$\sum_{n_1=0}^{\infty}((-1)^{n_1}*x_2^{2*n_1+1}*factorial(2*n_1+1)^{-1}*ln(n_0)^{2*n_1+1})$
$)$
$-(sin((q_2+2/4)*2\pi)*(1-x_1)+sin((q_2+1/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}$
$)$
$))))$

asendal naturallogaritmi summaga:

$\forall_{q_1}(\forall_{q_2}(\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\to$
$\zeta(x_1+i*x_2)=$
$\lim_{M\to\infty}($
$\sum_{n_0=1}^{floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})}(n_0{}^{-x_1}*$
$\sum_{n_1=0}^{\infty}((-1)^{n_1}*x_2^{2*n_1}*factorial(2*n_1)^{-1}*(\sum_{n_2=0}^{\infty}((2*n_2+1)^{-1}*(n_0-1)^{2*n_2+1}*(n_0+1)^{-2*n_2-1}))^{2*n_1}*2^{2*n_1})$
$)$
$-(sin((q_1+1/4)*2\pi)*(1-x_1)+sin((q_1+0/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}$
$)$
$+i*\lim_{M\to\infty}($
$\sum_{n_0=1}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})}(n_0{}^{-x_1}*\sum_{n_1=0}^{\infty}((\sum_{n_2=0}^{\infty}((2*n_2+1)^{-1}*(n_0-1)^{2*n_2+1}*(n_0+1)^{-2*n_2-1}))^{2*n_1+1}*x_2^{2*n_1+1}*$
$factorial(2*n_1+1)^{-1}*2^{2*n_1+1}*(-1)^{n_1})$
$)$
$-(sin((q_2+2/4)*2\pi)*(1-x_1)+sin((q_2+1/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}$
$)$
$))))$

kasutan multinomial valemit:

$\forall_{q_1}(\forall_{q_2}(\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\to$
$\zeta(x_1+i*x_2)=$
$\lim_{M\to\infty}($
$\sum_{n_0=1}^{floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})}(n_0{}^{-x_1}*\sum_{n_1=0}^{M-1}(x_2^{2*n_1}*2^{2*n_1}*(-1)^{n_1}*\sum_{n_2=0}^{(2*n_1+1)^M-1}($
$(\sum_{n_3=0}^{M-1}(\ floor(n_2/(2*n_1+1)^{n_3})\%(2*n_1+1)\ )=2*n_1)*$
$\Pi_{n_3=0}^{M-1}(factorial(\ floor(n_2/(2*n_1+1)^{n_3})\%(2*n_1+1)\ )^{-1}*((2*n_3+1)^{-1}*(n_0-1)^{2*n_3+1}*(n_0+1)^{-2*n_3-1})^{floor(n_2/(2*n_1+1)^{n_3})\%(2*n_1+1)}$
$)))$
$-(sin((q_1+1/4)*2\pi)*(1-x_1)+sin((q_1+0/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}$
$)$
$+i*\lim_{M\to\infty}($
$\sum_{n_0=1}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})}(n_0{}^{-x_1}*\sum_{n_1=0}^{M-1}(x_2^{2*n_1+1}*2^{2*n_1+1}*(-1)^{n_1}*\sum_{n_2=0}^{(2*n_1+2)^M-1}($
$(\sum_{n_3=0}^{M-1}(\ floor(n_2/(2*n_1+2)^{n_3})\%(2*n_1+2)\ )=2*n_1+1)*$
$\Pi_{n_3=0}^{M-1}(factorial(\ floor(n_2/(2*n_1+2)^{n_3})\%(2*n_1+2)\ )^{-1}*((2*n_3+1)^{-1}*(n_0-1)^{2*n_3+1}*(n_0+1)^{-2*n_3-1})^{floor(n_2/(2*n_1+2)^{n_3})\%(2*n_1+2)}$
$)))$
$-(sin((q_2+2/4)*2\pi)*(1-x_1)+sin((q_2+1/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}$
$)$
$))))$

lihtsustan teades, et korutamise asemel võib astmd liita.

$\forall_{q_1}(\forall_{q_2}(\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\to$
$\zeta(x_1+i*x_2)=$
$\lim_{M\to\infty}($

$\sum_{n_0=1}^{floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})}(n_0{}^{-x_1}*\sum_{n_1=0}^{M-1}(x_2^{2*n_1}*2^{2*n_1}*(-1)^{n_1}*\sum_{n_2=0}^{(2*n_1+1)^M-1}($
$(\sum_{n_3=0}^{M-1}(\ floor(n_2/(2*n_1+1)^{n_3})\%(2*n_1+1)\ )=2*n_1)*$
$(n_0-1)^{\sum_{n_3=0}^{M-1}((2*n_3+1)*(floor(n_2/(2*n_1+1)^{n_3})\%(2*n_1+1)))}*$
$(n_0+1)^{-\sum_{n_3=0}^{M-1}((2*n_3+1)*(floor(n_2/(2*n_1+1)^{n_3})\%(2*n_1+1)))}*$
$\Pi_{n_3=0}^{M-1}(factorial(\ floor(n_2/(2*n_1+1)^{n_3})\%(2*n_1+1)\ )^{-1}*(2*n_3+1)^{-floor(n_2/(2*n_1+1)^{n_3})\%(2*n_1+1)})$
$))$
$)))$
$-(\sin((q_1+1/4)*2\pi)*(1-x_1)+sin((q_1+0/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}$
$)$
$+i*\lim_{M\to\infty}($
$\sum_{n_0=1}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})}(n_0{}^{-x_1}*\sum_{n_1=0}^{M-1}(x_2^{2*n_1+1}*2^{2*n_1+1}*(-1)^{n_1}*\sum_{n_2=0}^{(2*n_1+2)^M-1}($
$(\sum_{n_3=0}^{M-1}(\ floor(n_2/(2*n_1+2)^{n_3})\%(2*n_1+2)\ )=2*n_1+1)*$
$(n_0-1)^{\sum_{n_3=0}^{M-1}((2*n_3+1)*(floor(n_2/(2*n_1+2)^{n_3})\%(2*n_1+2)))}*$
$(n_0+1)^{-\sum_{n_3=0}^{M-1}((2*n_3+1)*(floor(n_2/(2*n_1+2)^{n_3})\%(2*n_1+2)))}*$
$\Pi_{n_3=0}^{M-1}(factorial(\ floor(n_2/(2*n_1+2)^{n_3})\%(2*n_1+2)\ )^{-1}*(2*n_3+1)^{-floor(n_2/(2*n_1+2)^{n_3})\%(2*n_1+2)})$
$))$
$)))$
$-(\sin((q_2+2/4)*2\pi)*(1-x_1)+sin((q_2+1/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}$
$)$
$))))$

selle asemel et üle erinevate n2'ede ristsummade summeeridaarvutan lihtsalt n2'e ristsumma:

SIIN VIGA!

$\forall_{q_1}(\forall_{q_2}(\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\to$
$\zeta(x_1+i*x_2)=$
$\lim_{M\to\infty}($
$\sum_{n_0=1}^{floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})}(n_0^{-x_1}*\sum_{n_1=0}^{(2*M)^M-1}($
$(\sum_{n_2=0}^{M-1}(floor(n_1/(2*M)^{n_2})\%(2*M))\%2)*$
$(\sum_{k_2=0}^{M-1}(floor(k_1/(2*M-1)^{k_2})\%(2*M-1))<2*M-1)*$
$(y*2)^{\sum_{n_2=0}^{M-1}(floor(n_1/(2*M)^{n_2})\%(2*M))}*$
$(-1)^{\sum_{n_2=0}^{M-1}(floor(n_1/(2*M)^{n_2})\%(2*M))/2}*$
$(x-1)^{\sum_{n_2=0}^{M-1}((2*n_2+1)*(floor(n_1/(2*M)^{n_2})\%(2*M)))}*$
$(x+1)^{-\sum_{n_2=0}^{M-1}((2*n_2+1)*(floor(n_1/(2*M)^{n_2})\%(2*M)))}*$
$\Pi_{n_2=0}^{M-1}(factorial(floor(n_1/(2*M)^{n_2})\%(2*M))^{-1}*(2*n_2+1)^{-floor(n_1/(2*M)^{n_2})\%(2*M)})$
$))$
$-(sin((q_1+1/4)*2\pi)*(1-x_1)+sin((q_1+0/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}$
$)$
$+i*\lim_{M\to\infty}($
$\sum_{n_0=1}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})}(n_0^{-x_1}*\sum_{n_1=0}^{(2*M)^M-1}($
$(\sum_{n_2=0}^{M-1}(floor(n_1/(2*M)^{n_2})\%(2*M))\%2)*$
$(\sum_{n_2=0}^{M-1}(floor(n_1/(2*M)^{n_2})\%(2*M))<2*M)*$
$(x_2*2)^{\sum_{n_2=0}^{M-1}(floor(n_1/(2*M)^{n_2})\%(2*M))}*$
$(-1)^{(\sum_{n_2=0}^{M-1}(floor(n_1/(2*M)^{n_2})\%(2*M))+3)/2}*$
$(n_0-1)^{\sum_{n_2=0}^{M-1}((2*n_2+1)*(floor(n_1/(2*M)^{n_2})\%(2*M)))}*$
$(n_0+1)^{-\sum_{n_2=0}^{M-1}((2*n_2+1)*(floor(n_1/(2*M)^{n_2})\%(2*M)))}*$
$\Pi_{n_2=0}^{M-1}(factorial(floor(n_1/(2*M)^{n_2})\%(2*M))^{-1}*(2*n_2+1)^{-floor(n_1/(2*M)^{n_2})\%(2*M)})$
$))$

$$-(sin((q_2+2/4)*2\pi)*(1-x_1)+sin((q_2+1/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}$$
$$)$$
$$))))$$

kuna Mi lähenedes lõpmatusele see avaldis nagu nii koondub ja selle muutuse tegemine ainult lisab liikmeid, mis on absoluutväärtuselt väiksemad kui oleksid liikmed, mis lisanduksid Mi suurendamisel:

$$\forall_{q_1}(\forall_{q_2}(\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\rightarrow$$
$$\zeta(x_1+i*x_2)=$$
$$\lim_{M\rightarrow\infty}($$
$$\sum_{n_0=1}^{floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})}(n_0^{-x_1}*\sum_{n_1=0}^{(2*M)^M-1}($$
$$(\sum_{n_2=0}^{M-1}((floor(n_1/(2*M-1)^{n_2})\%(2*M-1))+1)\%2)*$$
$$(y*2)^{\sum_{n_2=0}^{M-1}(floor(n_1/(2*M-1)^{n_2})\%(2*M-1))}*$$
$$(-1)^{\sum_{n_2=0}^{M-1}(floor(n_1/(2*M-1)^{n_2})\%(2*M-1))/2}*$$
$$(n_0-1)^{\sum_{n_2=0}^{M-1}((2*n_2+1)*(floor(n_1/(2*M-1)^{n_2})\%(2*M-1)))}*$$
$$(n_0+1)^{-\sum_{n_2=0}^{M-1}((2*n_2+1)*(floor(n_1/(2*M-1)^{n_2})\%(2*M-1)))}*$$
$$\Pi_{n_2=0}^{M-1}(factorial(floor(n_1/(2*M-1)^{n_2})\%(2*M-1))^{-1}*(2*n_2+1)^{-floor(n_1/(2*M-1)^{n_2})\%(2*M-1)})$$
$$))/2$$
$$-(sin((q_1+1/4)*2\pi)*(1-x_1)+sin((q_1+0/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}$$
$$)$$
$$+i*\lim_{M\rightarrow\infty}($$
$$\sum_{n_0=1}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})}(n_0^{-x_1}*\sum_{n_1=0}^{(2*M)^M-1}($$
$$(\sum_{n_2=0}^{M-1}(floor(n_1/(2*M)^{n_2})\%(2*M))\%2)*$$
$$(x_2*2)^{\sum_{n_2=0}^{M-1}(floor(n_1/(2*M)^{n_2})\%(2*M))}*$$
$$(-1)^{(\sum_{n_2=0}^{M-1}(floor(n_1/(2*M)^{n_2})\%(2*M))+3)/2}*$$
$$(n_0-1)^{\sum_{n_2=0}^{M-1}((2*n_2+1)*(floor(n_1/(2*M)^{n_2})\%(2*M)))}*$$
$$(n_0+1)^{-\sum_{n_2=0}^{M-1}((2*n_2+1)*(floor(n_1/(2*M)^{n_2})\%(2*M)))}*$$
$$\Pi_{n_2=0}^{M-1}(factorial(floor(n_1/(2*M)^{n_2})\%(2*M))^{-1}*(2*n_2+1)^{-floor(n_1/(2*M)^{n_2})\%(2*M)})$$
$$))$$
$$-(sin((q_2+2/4)*2\pi)*(1-x_1)+sin((q_2+1/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}$$
$$)$$
$$))))$$

viin $n_0^{-x_1}$ summa sisse ja panen n_0 summa 0 ist algama.

$$\forall_{q_1}(\forall_{q_2}(\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\rightarrow$$
$$\zeta(x_1+i*x_2)=$$
$$\lim_{M\rightarrow\infty}($$
$$\sum_{n_0=0}^{floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})-1}(\sum_{n_1=0}^{(2*M)^M-1}($$
$$((1+\%2)*$$
$$(x_2*2)^{\sum_{n_2=0}^{M-1}(floor(n_1/(2*M)^{n_2})\%(2*M))}*$$
$$(-1)^{\sum_{n_2=0}^{M-1}(floor(n_1/(2*M)^{n_2})\%(2*M))/2}*$$
$$n_0^{\sum_{n_2=0}^{M-1}((2*n_2+1)*(floor(n_1/(2*M)^{n_2})\%(2*M)))}*$$
$$(n_0+1)^{-x_1}*$$
$$(n_0+2)^{-\sum_{n_2=0}^{M-1}((2*n_2+1)*(floor(n_1/(2*M)^{n_2})\%(2*M)))}*$$
$$\Pi_{n_2=0}^{M-1}(factorial(floor(n_1/(2*M)^{n_2})\%(2*M))^{-1}*(2*n_2+1)^{-floor(n_1/(2*M)^{n_2})\%(2*M)})$$
$$))$$
$$-(sin((q_1+1/4)*2\pi)*(1-x_1)+sin((q_1+0/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}$$
$$)$$
$$+i*\lim_{M\rightarrow\infty}($$
$$\sum_{n_0=0}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})-1}(\sum_{n_1=0}^{(2*M)^M-1}($$

$(\sum_{n_2=0}^{M-1}(floor(n_1/(2*M)^{n_2})\%(2*M))\%2)*$
$(x_2*2)^{\sum_{n_2=0}^{M-1}(floor(n_1/(2*M)^{n_2})\%(2*M))}*$
$(-1)^{(\sum_{n_2=0}^{M-1}(floor(n_1/(2*M)^{n_2})\%(2*M))+3)/2}*$
$n_0^{\sum_{n_2=0}^{M-1}((2*n_2+1)*(floor(n_1/(2*M)^{n_2})\%(2*M)))}*$
$(n_0+1)^{-x_1}*$
$(n_0+2)^{-\sum_{n_2=0}^{M-1}((2*n_2+1)*(floor(n_1/(2*M)^{n_2})\%(2*M)))}*$
$\Pi_{n_2=0}^{M-1}(factorial(floor(n_1/(2*M)^{n_2})\%(2*M))^{-1}*(2*n_2+1)^{-floor(n_1/(2*M)^{n_2})\%(2*M)})$
$))$
$-(sin((q_2+2/4)*2\pi)*(1-x_1)+sin((q_2+1/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}$
$)$
$))))$

lihtsustan viies n0-summa n1-summa sisse. ja muudan summade indekseid.

$\forall_{q_1}(\forall_{q_2}(\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\rightarrow$
$\zeta(x_1+i*x_2)=$
$\lim_{M\rightarrow\infty}(\sum_{n_0=0}^{(2*M)^M-1}($
$(\sum_{n_1=0}^{M-1}(1+floor(n_0/(2*M)^{n_1})\%(2*M))\%2)*$
$(x_2*2)^{\sum_{n_1=0}^{M-1}(floor(n_0/(2*M)^{n_1})\%(2*M))}*$
$(-1)^{\sum_{n_1=0}^{M-1}(floor(n_0/(2*M)^{n_1})\%(2*M))/2}*$
$\sum_{n_1=0}^{floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})-1}($
$n_1^{\sum_{n_2=0}^{M-1}((2*n_2+1)*(floor(n_0/(2*M)^{n_2})\%(2*M)))}*$
$(n_1+1)^{-x_1}*$
$(n_1+2)^{-\sum_{n_2=0}^{M-1}((2*n_2+1)*(floor(n_0/(2*M)^{n_2})\%(2*M)))}*$
$)$
$\Pi_{n_1=0}^{M-1}(factorial(floor(n_0/(2*M)^{n_1})\%(2*M))^{-1}*(2*n_1+1)^{-floor(n_0/(2*M)^{n_1})\%(2*M)})$
$)$
$-(sin((q_1+1/4)*2\pi)*(1-x_1)+sin((q_1+0/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}$
$)$
$+i*\lim_{M\rightarrow\infty}($
$\sum_{n_0=0}^{(2*M)^M-1}($
$(\sum_{n_1=0}^{M-1}(floor(n_0/(2*M)^{n_1})\%(2*M))\%2)*$
$(x_2*2)^{\sum_{n_1=0}^{M-1}(floor(n_0/(2*M)^{n_1})\%(2*M))}*$
$(-1)^{(\sum_{n_1=0}^{M-1}(floor(n_0/(2*M)^{n_1})\%(2*M))+3)/2}*$
$\sum_{n_1=0}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})-1}($
$n_1^{\sum_{n_2=2}^{M-1}((2*n_2+1)*(floor(n_0/(2*M)^{n_2})\%(2*M)))}*$
$(n_1+1)^{-x_1}*$
$(n_1+2)^{-\sum_{n_2=0}^{M-1}((2*n_2+1)*(floor(n_0/(2*M)^{n_2})\%(2*M)))}*$
$)$
$\Pi_{n_1=0}^{M-1}(factorial(floor(n_0/(2*M)^{n_1})\%(2*M))^{-1}*(2*n_1+1)^{-floor(n_0/(2*M)^{n_1})\%(2*M)})$
$)$
$-(sin((q_2+2/4)*2\pi)*(1-x_1)+sin((q_2+1/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}$
$)$
$))))$

lihtsustan n1-summa(vana n0-summa) sees olevat osa:

$\forall_{q_1}(\forall_{q_2}(\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\rightarrow$
$\zeta(x_1+i*x_2)=$
$\lim_{M\rightarrow\infty}(\sum_{n_0=0}^{(2*M)^M-1}($

$$(\sum_{n_1=0}^{M-1}(1+floor(n_0/(2*M)^{n_1})\%(2*M))\%2)*$$
$$(x_2*2)^{\sum_{n_1=0}^{M-1}(floor(n_0/(2*M)^{n_1})\%(2*M))}*$$
$$(-1)^{\sum_{n_1=0}^{M-1}(floor(n_0/(2*M)^{n_1})\%(2*M))/2}*$$
$$\sum_{n_1=2}^{floor(e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi})+1}($$
$$(1-2/n_1)^{\sum_{n_2=0}^{M-1}((2*n_2+1)*(floor(n_0/(2*M)^{n_2})\%(2*M)))}*$$
$$(n_1-1)^{-x_1}*$$
$$)$$
$$\Pi_{n_1=0}^{M-1}(factorial(floor(n_0/(2*M)^{n_1})\%(2*M))^{-1}*(2*n_1+1)^{-floor(n_0/(2*M)^{n_1})\%(2*M)})$$
$$)$$
$$-(sin((q_1+1/4)*2\pi)*(1-x_1)+sin((q_1+0/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q_1/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}$$
$$)$$
$$+i*\lim_{M\to\infty}($$
$$\sum_{n_0=0}^{(2*M)^M-1}($$
$$(\sum_{n_1=0}^{M-1}(floor(n_0/(2*M)^{n_1})\%(2*M))\%2)*$$
$$(x_2*2)^{\sum_{n_1=0}^{M-1}(floor(n_0/(2*M)^{n_1})\%(2*M))}*$$
$$(-1)^{(\sum_{n_1=0}^{M-1}(floor(n_0/(2*M)^{n_1})\%(2*M))+3)/2}*$$
$$\sum_{n_1=2}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})+1}($$
$$(1-2/n_1)^{\sum_{n_2=2}^{M-1}((2*n_2+1)*(floor(n_0/(2*M)^{n_2})\%(2*M)))}*$$
$$(n_1-1)^{-x_1}*$$
$$)$$
$$\Pi_{n_1=0}^{M-1}(factorial(floor(n_0/(2*M)^{n_1})\%(2*M))^{-1}*(2*n_1+1)^{-floor(n_0/(2*M)^{n_1})\%(2*M)})$$
$$)$$
$$-(sin((q_2+2/4)*2\pi)*(1-x_1)+sin((q_2+1/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}$$
$$)$$
$$))))$$

1.2.3 tagumine pool nulliks

$$sin((q_1+1/4)*2\pi)*(1-x_1)+sin((q_1+0/4)*2\pi)*x_2=0\wedge$$
$$sin((q_2+2/4)*2\pi)*(1-x_1)+sin((q_2+1/4)*2\pi)*x_2=0$$

$$\mathrm{sin}((\mathrm{q}_1+1/4)*2\pi)*(1-x_1)+sin(q_1*2\pi)*x_2=0\wedge$$
$$sin((q_2+2/4)*2\pi)*(1-x_1)+sin((q_2+1/4)*2\pi)*x_2=0$$

$$\exists_k(k\in N\wedge q_1=k/2-arctan((1-x_1)/x_2)/(2\pi))$$
$$\exists_k(k\in N\wedge q_2=k/2+arctan(x_2/(1-x_1))/(2\pi))$$

$$\exists_k(k\in N\wedge q_2=k/2-arctan((1-x_1)/x_2)/(2\pi)+1/4)$$

$$arctan(x_2/(1-x_1))+arctan((1-x_1)/x_2)=2\pi/4*sgn((1-x_1)*x_2)$$
$$arctan(x_2/(1-x_1))=-arctan((1-x_1)/x_2)+2\pi/4*sgn((1-x_1)*x_2)$$

$$\exists_k(k\in N\wedge q_2=k/2-arctan((1-x_1)/x_2)/(2\pi)+1/4*sgn((1-x_1)*x_2))$$

$$\mathrm{q}_2=q_1+1/4*sgn((1-x_1)*x_2))+k/2$$
$$\mathrm{q}_2=q_1+1/4$$
$$\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\rightarrow$$
$$\zeta(x_1+i*x_2)=$$

$\lim_{M\to\infty}(\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+(k/2-arctan((1-x_1)/x_2)/(2\pi))/x_2)*2\pi})}(sin((x_2*ln(n)/(2\pi)+1/4)*2\pi)*n^{-x_1}))$
$+i*\lim_{M\to\infty}(-\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+(k/2+arctan(x_2/(1-x_1))/(2\pi))/x_2)*2\pi})}(sin((x_2*ln(n)/(2\pi))*2\pi)*n^{-x_1}))=$

$\lim_{M\to\infty}(\sum_{n=1}^{floor(e^{(floor(M)*2\pi/abs(x_2)+k*2\pi/2-arctan((1-x_1)/x_2)/x_2)})}(sin(x_2*ln(n)+2\pi/4)*n^{-x_1}))$
$+i*\lim_{M\to\infty}(-\sum_{n=1}^{floor(e^{(floor(M)*2\pi/abs(x_2)+k*2\pi/2+arctan(x_2/(1-x_1))/x_2)})}(sin(x_2*ln(n))*n^{-x_1}))=$

$\lim_{M\to\infty}(\sum_{n=1}^{floor(e^{(floor(M)*2\pi/2/abs(x_2)-arctan((1-x_1)/x_2)/x_2)})}(sin(x_2*ln(n)+2\pi/4)*n^{-x_1}))$
$+i*\lim_{M\to\infty}(-\sum_{n=1}^{floor(e^{(floor(M)*2\pi/2/abs(x_2)+arctan(x_2/(1-x_1))/x_2)})}(sin(x_2*ln(n))*n^{-x_1}))=$

$\lim_{M\to\infty}(\sum_{n=1}^{floor(e^{(floor(M)*2\pi/2/abs(x_2)-arctan((1-x_1)/x_2)/x_2)})}(sin(x_2*ln(n)+2\pi/4)*n^{-x_1}))$
$+i*\lim_{M\to\infty}(-\sum_{n=1}^{floor(e^{(floor(M)*2\pi/2/abs(x_2)+(-arctan((1-x_1)/x_2)/x_2+2\pi/4*sgn((1-x_1)*x_2))/x_2)})}(sin(x_2*ln(n))*n^{-x_1}))=$

$\lim_{M\to\infty}(\sum_{n=1}^{floor(e^{floor(M)*2\pi/2/abs(x_2)-arctan((1-x_1)/x_2)/x_2})}(sin(x_2*ln(n)+2\pi/4)*n^{-x_1}))$
$+i*\lim_{M\to\infty}(-\sum_{n=1}^{floor(e^{floor(M)*2\pi/2/abs(x_2)-arctan((1-x_1)/x_2)/x_2+2\pi/4*sgn((1-x_1)*x_2)/x_2)})}(sin(x_2*ln(n))*n^{-x_1}))=$

$))$

Valem zeta13_tagumine_pool_nulliks:

$\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\to$
$\zeta(x_1+i*x_2)=$
$\lim_{M\to\infty}(\sum_{n=1}^{floor(e^{floor(M)/2*2\pi/abs(x_2)-arctan((1-x_1)/x_2)/x_2})}(sin(x_2*ln(n)+2\pi/4)*n^{-x_1}))$
$+i*\lim_{M\to\infty}(-\sum_{n=1}^{floor(e^{(floor(M)/2+1/4)*2\pi/abs(x_2)-arctan((1-x_1)/x_2)/x_2})}(sin(x_2*ln(n))*n^{-x_1}))$
$))$

asendan siinuse ja logaritmi vastavate Taylori ridadega.

$\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\to$
$\zeta(x_1+i*x_2)=$
$\lim_{M\to\infty}(\sum_{n_0=1}^{floor(e^{floor(M)/2*2\pi/abs(x_2)-arctan((1-x_1)/x_2)/x_2})}($
$\sum_{n_1=0}^{M}($
$(-1)^{n_1}*(fact(2*n_1))^{-1}*(x_2*ln(n_0)+2\pi/4)^{2*n_1}*n_0^{-x_1})))$
$+i*\lim_{M\to\infty}(-\sum_{n_0=1}^{floor(e^{(floor(M)/2+1/4)*2\pi/abs(x_2)-arctan((1-x_1)/x_2)/x_2})}(sin(x_2*ln(n_0))*n_0^{-x_1}))$
$))$

asendan logaritmi vastavate Taylori ridadega.

valem zeta14_tagumine_pool_nulliks_tayloriga(ei suutnud kontrollida, aga vb sest arvutusvõime piiratuse tõttu ei saanud piidsavaölt suurt Mi vsalida):

$\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\to$
$\zeta(x_1+i*x_2)=$
$\lim_{M\to\infty}(\sum_{n_0=0}^{floor(e^{floor(M)/2*2\pi/abs(x_2)-arctan((1-x_1)/x_2)/x_2})}(\sum_{n_1=0}^{(2*M)^M-1}($
$(1+\%(2*M))\%2)*$
$(x_2*2)^{\sum_{n_2=0}^{M-1}(floor(n_1/(2*M)^{n_2})\%(2*M))}*$
$(-1)^{\sum_{n_2=0}^{M-1}(floor(n_1/(2*M)^{n_2})\%(2*M))/2}*$
$n_0^{\sum_{n_2=0}^{M-1}((2*n_2+1)*(floor(n_1/(2*M)^{n_2})\%(2*M)))}*$
$(n_0+1)^{-x_1}*$
$(n_0+2)^{-\sum_{n_2=0}^{M-1}((2*n_2+1)*(floor(n_1/(2*M)^{n_2})\%(2*M)))}*$
$\Pi_{n_2=0}^{M-1}(factorial(floor(n_1/(2*M)^{n_2})\%(2*M))^{-1}*(2*n_2+1)^{-floor(n_1/(2*M)^{n_2})\%(2*M)})$
$)))$

$+i*\lim_{M\to\infty}(\sum_{n_0=0}^{floor(e^{(floor(M)/2+1/4)*2\pi/abs(x_2)-arctan((1-x_1)/x_2)/x_2})}(\sum_{n_1=0}^{(2*M)^M-1}($
$\sum_{n_2=0}^{M-1}(floor(n_1/(2*M)^{n_2})\%(2*M))\%2)*$

$(x_2*2)^{\sum_{n_2=0}^{M-1}(floor(n_1/(2*M)^{n_2})\%(2*M))}*$
$(-1)^{(\sum_{n_2=0}^{M-1}(floor(n_1/(2*M)^{n_2})\%(2*M))+1)/2}*$
$n_0^{\sum_{n_2=0}^{M-1}((2*n_2+1)*(floor(n_1/(2*M)^{n_2})\%(2*M)))}*$
$(n_0+1)^{-x_1}*$
$(n_0+2)^{-\sum_{n_2=0}^{M-1}((2*n_2+1)*(floor(n_1/(2*M)^{n_2})\%(2*M)))}*$
$\Pi_{n_2=0}^{M-1}(factorial(floor(n_1/(2*M)^{n_2})\%(2*M))^{-1}*(2*n_2+1)^{-floor(n_1/(2*M)^{n_2})\%(2*M)})$
)))

))

1.2.4 $q_1 = q_2$

$\forall_{q_1}(\forall_{x_1}(\forall_{x_2}(x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \rightarrow$

$\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+q/x_2)*2\pi})}(sin(x_2*ln(n)+(2\pi)*1/4)*n^{-x_1})$
$-(sin((q+1/4)*2\pi)*(1-x_1)+sin((q+0/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q/x_2)*2\pi*(1-x_1)/x_2}*((1-x_1)^2+x_2^2)^{-1}$
$)$
$+i*\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+q/x_2)*2\pi})}(sin(x_2*ln(n)+(2\pi)*2/4)*n^{-x_1})$
$-(sin((q+2/4)*2\pi)*(1-x_1)+sin((q+1/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+q/x_2)*2\pi*(1-x_1)/x_2}*((1-x_1)^2+x_2^2)^{-1}$
$)$
))))

$q_1 = 0 \wedge q_2 = 0$ kui valida, et $q_1 = 0$ ja $q_2 = 0$. Valem zeta12:
$\forall_{x_1}(\forall_{x_2}(x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \rightarrow$
$\zeta(x_1+i*x_2) =$

$\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{floor(M)*2\pi/abs(x_2)})}(sin(x_2*ln(n)+1*(2\pi)/4)*n^{-x_1})$
$-e^{floor(M)*2\pi*(1-x_1)/abs(x_2)}*((1-x_1)^2+x_2^2)^{-1}*(1-x_1)$
$)$
$+i*\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{floor(M)*2\pi/abs(x_2)})}(sin(x_2*ln(n)+2*(2\pi)/4)*n^{-x_1})$
$-e^{floor(M)*2\pi*(1-x_1)/abs(x_2)}*((1-x_1)^2+x_2^2)^{-1}*x_2$
$)=$
))

1.2.5 $q_1 = 1/8 \wedge q_2 = -1/8$

$\forall_{x_1}(\forall_{x_2}(x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \rightarrow$
$\zeta(x_1+i*x_2) =$

$\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+1/8/x_2)*2\pi})}(sin((x_2*ln(n)/(2\pi)+1/4)*2\pi)*n^{-x_1})$
$-(sin((1/8+1/4)*2\pi)*(1-x_1)+sin((1/8+0/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)+1/8/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}$
$)$
$+i*\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)-1/8/x_2)*2\pi})}(sin((x_2*ln(n)/(2\pi)+2/4)*2\pi)*n^{-x_1})$
$-(sin((-1/8+2/4)*2\pi)*(1-x_1)+sin((-1/8+1/4)*2\pi)*x_2)*e^{(floor(M)/abs(x_2)-1/8/x_2)*2\pi*(1-x_1)/x_2}*((1-x_1)^2+$

$x_2^2)^{-1}$
$) =$

$\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)+1/8/x_2)*2\pi})}(sin((x_2*ln(n)/(2\pi)+1/4)*2\pi)*n^{-x_1})$
$-((1-x_1)+x_2)*e^{(floor(M)/abs(x_2)+1/8/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}*2^{-1/2}$
$)$
$+i*\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)/abs(x_2)-1/8/x_2)*2\pi})}(sin((x_2*ln(n)/(2\pi)+2/4)*2\pi)*n^{-x_1})$
$-((1-x_1)+x_2)*e^{(floor(M)/abs(x_2)-1/8/x_2)*2\pi*(1-x_1)}*((1-x_1)^2+x_2^2)^{-1}*2^{-1/2}$
$) =$

))

1.2.6 kirjutades siinuse ja koosiinuse summana

kasutatud valem $sin(x) = \sum_{m=0}^{\infty}(x^{2*m+1}*((2*m+1)!)^{-1}*(-1)^m)$
$cos(x) = \sum_{m=0}^{\infty}(x^{2*m}*((2*m)!)^{-1}*(-1)^m)$

zeta ise $\forall_{x_1}(\forall_{x_2}(x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \rightarrow$
$\zeta(x_1+i*x_2) =$
$(x_1^2-2*x_1+1+x_2^2)^{-1}*\sum_{n=1}^{\infty}($
$\sum_{m=0}^{\infty}(ln(n*(2*m+1))*x_2^{2*m+1}*((2*m+1)!)^{-1}*(-1)^m)*x_2*(n-1)*n^{-x_1}+$
$\sum_{m=0}^{\infty}(ln(n*2*m)*x_2^{2*m}*((2*m)!)^{-1}*(-1)^m)*(x_1^2-x_1-n*x_1+n+x_2^2)*n^{-x_1}-$
$\sum_{m=0}^{\infty}(ln((n+1)*(2*m+1))*x_2^{2*m+1}*((2*m+1)!)^{-1}*(-1)^m)*x_2*n*(n+1)^{-x_1}-$
$\sum_{m=0}^{\infty}(ln((n+1)*2*m)*x_2^{2*m}*((2*m)!)^{-1}*(-1)^m)*(1-x_1)*n*(n+1)^{-x_1}$
$)$
$+i*(x_1^2-2*x_1+1+x_2^2)^{-1}*\sum_{n=1}^{\infty}($
$(sin(x_2*ln(n)+1*(2\pi)/4)*x_2*(n-1)+sin(x_2*ln(n)+2*(2\pi)/4)*(x_1^2-x_1-n*x_1+n+x_2^2))*n^{-x_1}-$
$(sin(x_2*ln(n+1)+1*(2\pi)/4)*x_2+sin(x_2*ln(n+1)+2*(2\pi)/4)*(1-x_1))*n*(n+1)^{-x_1}$
$)$
$=$
$(x_1^2-2*x_1+1+x_2^2)^{-1}*\sum_{n=1}^{\infty}(\sum_{m=0}^{\infty}(x_2^{2*m}*(-1)^m*((2*m)!)^{-1}*($
$ln(n*(2*m+1))*(2*m+1)^{-1}*x_2^2*(n-1)*n^{-x_1}+$
$ln(n*2*m)*(x_1^2-x_1-n*x_1+n+x_2^2)*n^{-x_1}-$
$ln((n+1)*(2*m+1))*(2*m+1)^{-1}*x_2^2*n*(n+1)^{-x_1}-$
$ln((n+1)*2*m)*(1-x_1)*n*(n+1)^{-x_1}$
$)))$
$+i*(x_1^2-2*x_1+1+x_2^2)^{-1}*\sum_{n=1}^{\infty}($
$(sin(x_2*ln(n)+1*(2\pi)/4)*x_2*(n-1)+sin(x_2*ln(n)+2*(2\pi)/4)*(x_1^2-x_1-n*x_1+n+x_2^2))*n^{-x_1}-$
$(sin(x_2*ln(n+1)+1*(2\pi)/4)*x_2+sin(x_2*ln(n+1)+2*(2\pi)/4)*(1-x_1))*n*(n+1)^{-x_1}$
$)$
$=))$

2 zetafunktsiooni nullkohad

Võrdsustan zetafunktsiooni kuju zeta_2_q_ga, mille mujal tuletasin 0iga.
$\forall_{q_1}(\forall_{q_2}(\forall_{x_1}(\forall_{x_2}(x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \wedge \zeta(x_1+i*x_2) = 0 \rightarrow$

$\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)+q_1)*2\pi/x_2})}(sin(x_2*ln(n)+(2\pi)*1/4)*n^{-x_1})$
$-(sin((q_1+1/4)*2\pi)*(1-x_1)+sin((q_1+0/4)*2\pi)*x_2)*e^{(floor(M)+q_1)*2\pi*(1-x_1)/x_2}*((1-x_1)^2+x_2^2)^{-1}$

$$
)
+i*\lim_{M\to\infty}(
\sum_{n=1}^{floor(e^{(floor(M)+q_2)*2\pi/x_2})}(sin(x_2*ln(n)+(2\pi)*2/4)*n^{-x_1})
-(sin((q_2+2/4)*2\pi)*(1-x_1)+sin((q_2+1/4)*2\pi)*x_2)*e^{(floor(M)+q_2)*2\pi*(1-x_1)/x_2}*((1-x_1)^2+x_2^2)^{-1}
)=0
))))
$$

kuna nii reaalosa kui imaginaarosa peavad nullid olema, siis võin imaginnarosa suvalise konstandiga läbi korrutada.

$$
\forall_{q_1}(\forall_{q_2}(\forall_{Q_1}(\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\wedge\zeta(x_1+i*x_2)=0\to
$$

$$
Q_1*\lim_{M\to\infty}(
\sum_{n=1}^{floor(e^{(floor(M)+q_1)*2\pi/x_2})}(sin(x_2*ln(n)+(2\pi)*1/4)*n^{-x_1})
-(sin((q_1+1/4)*2\pi)*(1-x_1)+sin((q_1+0/4)*2\pi)*x_2)*e^{(floor(M)+q_1)*2\pi*(1-x_1)/x_2}*((1-x_1)^2+x_2^2)^{-1}=0
)+
\lim_{M\to\infty}(
\sum_{n=1}^{floor(e^{(floor(M)+q_2)*2\pi/x_2})}(sin(x_2*ln(n)+(2\pi)*2/4)*n^{-x_1})
-(sin((q_2+2/4)*2\pi)*(1-x_1)+sin((q_2+1/4)*2\pi)*x_2)*e^{(floor(M)+q_2)*2\pi*(1-x_1)/x_2}*((1-x_1)^2+x_2^2)^{-1}
)=0
))))
$$

viin konsatndi sulguse sisse.

$$
\forall_{q_1}(\forall_{q_2}(\forall_{Q_1}(\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\wedge\zeta(x_1+i*x_2)=0\to
$$

$$
\lim_{M\to\infty}(
Q_1*\sum_{n=1}^{floor(e^{(floor(M)+q_1)*2\pi/x_2})}(sin(x_2*ln(n)+(2\pi)*1/4)*n^{-x_1})
-Q_1*(sin((q_1+1/4)*2\pi)*(1-x_1)+sin((q_1+0/4)*2\pi)*x_2)*e^{(floor(M)+q_1)*2\pi*(1-x_1)/x_2}*((1-x_1)^2+x_2^2)^{-1}+
\sum_{n=1}^{floor(e^{(floor(M)+q_2)*2\pi/x_2})}(sin(x_2*ln(n)+(2\pi)*2/4)*n^{-x_1})
-(sin((q_2+2/4)*2\pi)*(1-x_1)+sin((q_2+1/4)*2\pi)*x_2)*e^{(floor(M)+q_2)*2\pi*(1-x_1)/x_2}*((1-x_1)^2+x_2^2)^{-1}
)=0
))))
$$

muudan kirjutan ümber viies ühiseid kordajaid sulguse ette.

$$
\forall_{q_1}(\forall_{q_2}(\forall_{Q_1}(\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\wedge\zeta(x_1+i*x_2)=0\to
$$

$$
\lim_{M\to\infty}(
\sum_{n=1}^{floor(e^{(floor(M)+q_1)*2\pi/x_2})}(Q_1*sin(x_2*ln(n)+(2\pi)*1/4)*n^{-x_1}+sin(x_2*ln(n)+(2\pi)*2/4)*n^{-x_1})
-((Q_1*sin((q_1+1/4)*2\pi)+sin((q_2+2/4)*2\pi))*(1-x_1)+(Q_1*sin((q_1+0/4)*2\pi)+sin((q_2+1/4)*2\pi))*
x_2)*e^{(floor(M)+q_1)*2\pi*(1-x_1)/x_2}*((1-x_1)^2+x_2^2)^{-1})=0
))))
$$

siinuse ja koosinuse summa saab ühiseks siinuseks teha nii, et argumendile on midagi liidetud.

$$
\forall_{q_1}(\forall_{Q_1}(\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\wedge\zeta(x_1+i*x_2)=0\to
$$

$$
\lim_{M\to\infty}(
\sum_{n=1}^{floor(e^{(floor(M)+q_1)*2\pi/x_2})}(\sqrt{Q_1^2+1}*-sin(x_2*ln(n)+arctan(-Q_1))*n^{-x_1})
-((-\sqrt{Q_1^2+1}*sin(q_1*2\pi+arctan(-Q_1)))*(1-x_1)+\sqrt{Q_1^2+1}*sin(q_1*2\pi+arctan(1/Q_1))*x_2)
e^{(floor(M)+q_1)*2\pi*(1-x_1)/x_2}*((1-x_1)^2+x_2^2)^{-1})=0
)))
$$

arctan(-x)=arctan(x)

$$
\forall_{q_1}(\forall_{Q_1}(\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\wedge\zeta(x_1+i*x_2)=0\to
$$

$$
\lim_{M\to\infty}(
\sum_{n=1}^{floor(e^{(floor(M)+q_1)*2\pi/x_2})}(-sin(x_2*ln(n)-arctan(Q_1))*n^{-x_1})
$$

$-(-sin(q_1*2\pi-arctan(Q_1))*(1-x_1)+sin(q_1*2\pi+arctan(1/Q_1))*x_2)$
$e^{(floor(M)+q_1)*2\pi*(1-x_1)/x_2}*((1-x_1)^2+x_2^2)^{-1})$
$)=0$
$)))$

arctan(1/x)=-arctan(x)+sgn(x)*2\pi/4

$\forall_{q_1}(\forall_{Q_1}(\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\wedge\zeta(x_1+i*x_2)=0\rightarrow$

$lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)+q_1)*2\pi/x_2})}(-sin(x_2*ln(n)-arctan(Q_1))*n^{-x_1})$
$-(-sin(q_1*2\pi-arctan(Q_1))*(1-x_1)+sin(q_1*2\pi-arctan(Q_1)+sgn(Q_1)*2\pi/4)*x_2)$
$e^{(floor(M)+q_1)*2\pi*(1-x_1)/x_2}*((1-x_1)^2+x_2^2)^{-1})$
$)=0$
$)))$

korrutan -1ega läbi

$\forall_{q_1}(\forall_{Q_1}(\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\wedge\zeta(x_1+i*x_2)=0\rightarrow$

$lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)+q_1)*2\pi/x_2})}(sin(x_2*ln(n)-arctan(Q_1))*n^{-x_1})$
$-(sin(q_1*2\pi-arctan(Q_1))*(1-x_1)-sin(q_1*2\pi-arctan(Q_1)+sgn(Q_1)*2\pi/4)*x_2)$
$e^{(floor(M)+q_1)*2\pi*(1-x_1)/x_2}*((1-x_1)^2+x_2^2)^{-1})$
$)=0$
$)))$

2pi/4 lahutamine on sama kui 2pi/4 lahutamine ja siis -1'ega läbi korrutamine

$\forall_{q_1}(\forall_{Q_1}(\forall_{x_1}(\forall_{x_2}(x_1\in R\wedge x_2\in R\wedge x_1>0\wedge\zeta(x_1+i*x_2)=0\rightarrow$

$lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)+q_1)*2\pi/x_2})}(sin(x_2*ln(n)-arctan(Q_1))*n^{-x_1})$
$-(sin(q_1*2\pi-arctan(Q_1))*(1-x_1)-sin(q_1*2\pi-arctan(Q_1)+2\pi/4)*sgn(Q_1)*x_2)$
$e^{(floor(M)+q_1)*2\pi*(1-x_1)/x_2}*((1-x_1)^2+x_2^2)^{-1})$
$)=0$
$)))$

kui Q_1 on positiivne, siis sgn(Q_1)=1, kui negatiivne, siis sgn(Q_1)=-1

$\forall_{q_1}(\forall_{x_1}(\forall_{x_2}(\forall_{Q_1}(q_1\in R\wedge Q_1\in R\wedge x_1\in R\wedge x_2\in R\wedge x_1>0\wedge\zeta(x_1+i*x_2)=0\rightarrow($
$Q_1>0land($
$lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)+q_1)*2\pi/x_2})}(sin(x_2*ln(n)-arctan(Q_1))*n^{-x_1})$
$-(sin(q_1*2\pi-arctan(Q_1))*(1-x_1)-sin(q_1*2\pi-arctan(Q_1)+2\pi/4)*x_2)$
$e^{(floor(M)+q_1)*2\pi*(1-x_1)/x_2}*((1-x_1)^2+x_2^2)^{-1})$
$)=0$
$\vee$
$Q_1<0\wedge($
$lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)+q_1)*2\pi/x_2})}(sin(x_2*ln(n)-arctan(Q_1))*n^{-x_1})$
$-(sin(q_1*2\pi-arctan(Q_1))*(1-x_1)+sin(q_1*2\pi-arctan(Q_1)+2\pi/4)*x_2)$
$e^{(floor(M)+q_1)*2\pi*(1-x_1)/x_2}*((1-x_1)^2+x_2^2)^{-1})$
$)=0$
$))))$

arctan(Q_1) võtab väärtuseid -2pi/4 kuni 2pi/4

$\forall_{q_1}(\forall_{x_1}(\forall_{x_2}(\forall_{Q_1}(q_1\in R\wedge Q_1\in R\wedge x_1\in R\wedge x_2\in R\wedge x_1>0\wedge\zeta(x_1+i*x_2)=0\rightarrow($
$(Q_1>-2\pi/4\wedge Q_1<0)\wedge($
$lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)+q_1)*2\pi/x_2})}(sin(x_2*ln(n)-Q_1)*n^{-x_1})$

$-(sin(q_1*2\pi-Q_1)*(1-x_1)-sin(q_1*2\pi-Q_1+2\pi/4)*x_2)$
$e^{(floor(M)+q_1)*2\pi*(1-x_1)/x_2}*((1-x_1)^2+x_2^2)^{-1}$
$)=0\vee$

$(Q_1>0\wedge Q_1<2\pi/4)\wedge($
$\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)+q_1)*2\pi/x_2})}(sin(x_2*ln(n)-Q_1)*n^{-x_1})$
$-(sin(q_1*2\pi-Q_1)*(1-x_1)+sin(q_1*2\pi-Q_1+2\pi/4)*x_2)$
$e^{(floor(M)+q_1)*2\pi*(1-x_1)/x_2}*((1-x_1)^2+x_2^2)^{-1}$
$)=0$
$)$

))))

Ühes kohas Q_1 asemele samanimeline muutuja -Q_1. Teises kohas Q_1 asemele samanimeline muutuja 2\pi/2-Q_1.

$\forall_{q_1}(\forall_{x_1}(\forall_{x_2}(\forall_{Q_1}(q_1\in R\wedge Q_1\in R\wedge x_1\in R\wedge x_2\in R\wedge x_1>0\wedge\zeta(x_1+i*x_2)=0\to($
$(Q_1>0\wedge Q_1<2\pi/4)\wedge($
$\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)+q_1)*2\pi/x_2})}(sin(x_2*ln(n)+Q_1)*n^{-x_1})$
$-(sin(q_1*2\pi+Q_1)*(1-x_1)-sin(q_1*2\pi+Q_1+2\pi/4)*x_2)$
$e^{(floor(M)+q_1)*2\pi*(1-x_1)/x_2}*((1-x_1)^2+x_2^2)^{-1}$
$)=0\vee$

$(Q_1>2\pi/4\wedge Q_1<2\pi/2)\wedge($
$\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)+q_1)*2\pi/x_2})}(sin(x_2*ln(n)-2\pi/2+Q_1)*n^{-x_1})$
$-(sin(q_1*2\pi-2\pi/2+Q_1)*(1-x_1)+sin(q_1*2\pi-2\pi/2+Q_1+2\pi/4)*x_2)$
$e^{(floor(M)+q_1)*2\pi*(1-x_1)/x_2}*((1-x_1)^2+x_2^2)^{-1}$
$)=0$
$)$

))))

arvestan, et sin(x-2pi/2)=-sin(x)

$\forall_{q_1}(\forall_{x_1}(\forall_{x_2}(\forall_{Q_1}(q_1\in R\wedge Q_1\in R\wedge x_1\in R\wedge x_2\in R\wedge x_1>0\wedge\zeta(x_1+i*x_2)=0\to($
$(Q_1>0\wedge Q_1<2\pi/4)\wedge($
$\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)+q_1)*2\pi/x_2})}(sin(x_2*ln(n)+Q_1)*n^{-x_1})$
$-(sin(q_1*2\pi+Q_1)*(1-x_1)-sin(q_1*2\pi+Q_1+2\pi/4)*x_2)$
$e^{(floor(M)+q_1)*2\pi*(1-x_1)/x_2}*((1-x_1)^2+x_2^2)^{-1}$
$)=0\vee$

$(Q_1>2\pi/4\wedge Q_1<2\pi/2)\wedge($
$\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)+q_1)*2\pi/x_2})}(-sin(x_2*ln(n)+Q_1)*n^{-x_1})$
$-(-sin(q_1*2\pi+Q_1)*(1-x_1)-sin(q_1*2\pi+Q_1+2\pi/4)*x_2)$
$e^{(floor(M)+q_1)*2\pi*(1-x_1)/x_2}*((1-x_1)^2+x_2^2)^{-1}$
$)=0$
$)$

))))

Arvestan, et 0*(-1)=0 ja sin(x+2\pi/2)=-sin(x). Muutuja Q_1 asemele muuutuja Q_1-2\pi/2.

$\forall_{q_1}(\forall_{x_1}(\forall_{x_2}(\forall_{Q_1}(q_1\in R\wedge Q_1\in R\wedge x_1\in R\wedge x_2\in R\wedge x_1>0\wedge\zeta(x_1+i*x_2)=0\to($

$(Q_1 > 0 \wedge Q_1 < 2\pi/4) \wedge ($
$\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)+q_1)*2\pi/x_2})}(sin(x_2 * ln(n) + Q_1) * n^{-x_1})$
$-(sin(q_1 * 2\pi + Q_1) * (1-x_1) - sin(q_1 * 2\pi + Q_1 + 2\pi/4) * x_2)$
$e^{(floor(M)+q_1)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1}$
$) = 0$
$\vee$

$(Q_1 > 2\pi/4 \wedge Q_1 < 2\pi/2) \wedge ($
$\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)+q_1)*2\pi/x_2})}(-sin(x_2 * ln(n) + Q_1) * n^{-x_1})$
$-(-sin(q_1 * 2\pi + Q_1) * (1-x_1) - sin(q_1 * 2\pi + Q_1 + 2\pi/4) * x_2)$
$e^{(floor(M)+q_1)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1}$
$) = 0$
$\vee$

$(\mathrm{Q}_1 > 2\pi/2 \wedge Q_1 < 2\pi * 3/4) \wedge ($
$\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)+q_1)*2\pi/x_2})}(sin(x_2 * ln(n) + Q_1) * n^{-x_1})$
$-(sin(q_1 * 2\pi + Q_1) * (1-x_1) - sin(q_1 * 2\pi + Q_1 + 2\pi/4) * x_2)$
$e^{(floor(M)+q_1)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1}$
$) = 0$
$\vee$

$(Q_1 > 2\pi * 3/4 \wedge Q_1 < 2\pi) \wedge ($
$\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)+q_1)*2\pi/x_2})}(-sin(x_2 * ln(n) + Q_1) * n^{-x_1})$
$-(-sin(q_1 * 2\pi + Q_1) * (1-x_1) - sin(q_1 * 2\pi + Q_1 + 2\pi/4) * x_2)$
$e^{(floor(M)+q_1)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1}$
$) = 0$

)

))))
korrutan osad osad -1ega läbi. ja panaen osad kokku.
$\forall_{q_1}(\forall_{x_1}(\forall_{x_2}(\forall_{Q_1}(q_1 \in R \wedge Q_1 \in R \wedge x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \wedge \zeta(x_1 + i * x_2) = 0 \to ($
$(Q_1 > 0 \wedge Q_1 < 2\pi/4 \vee Q_1 > 2\pi/2 \wedge Q_1 < 2\pi * 3/4) \wedge ($
$\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)+q_1)*2\pi/x_2})}(sin(x_2 * ln(n) + Q_1) * n^{-x_1})$
$-(sin(q_1 * 2\pi + Q_1) * (1-x_1) - sin(q_1 * 2\pi + Q_1 + 2\pi/4) * x_2)$
$e^{(floor(M)+q_1)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1}$
$) = 0$
$\vee$

$(Q_1 > 2\pi/4 \wedge Q_1 < 2\pi/2 \vee Q_1 > 2\pi * 3/4 \wedge Q_1 < 2\pi) \wedge ($
$\lim_{M\to\infty}($
$\sum_{n=1}^{floor(e^{(floor(M)+q_1)*2\pi/x_2})}(sin(x_2 * ln(n) + Q_1) * n^{-x_1})$
$-(sin(q_1 * 2\pi + Q_1) * (1-x_1) + sin(q_1 * 2\pi + Q_1 + 2\pi/4) * x_2)$
$e^{(floor(M)+q_1)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1}$
$) = 0$

)

))))

üheks oskas kokku.

$$\forall_{q_1}(\forall_{x_1}(\forall_{x_2}(\forall_{Q_1}(q_1 \in R \wedge Q_1 \in R \wedge x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \wedge \zeta(x_1 + i * x_2) = 0 \rightarrow (
\lim_{M \to \infty}(
\sum_{n=1}^{floor(e^{(floor(M)+q_1)*2\pi/x_2})}(sin(x_2 * ln(n) + Q_1) * n^{-x_1})
-(sin(q_1 * 2\pi + Q_1) * (1 - x_1) - sin(q_1 * 2\pi + Q_1 + 2\pi/4) * sgn(Q_1 \% (2\pi/2) - 2\pi/4) * x_2)
e^{(floor(M)+q_1)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1}
) = 0
)
))))$$

muutuja Q_1 asemele Q_1/(2\pi)

$$\forall_{q_1}(\forall_{x_1}(\forall_{x_2}(\forall_{Q_1}(q_1 \in R \wedge Q_1 \in R \wedge x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \wedge \zeta(x_1 + i * x_2) = 0 \rightarrow (
\lim_{M \to \infty}(
\sum_{n=1}^{floor(e^{(floor(M)+q_1)*2\pi/x_2})}(sin(x_2 * ln(n) + Q_1 * 2\pi) * n^{-x_1})
-(sin((q_1 + Q_1) * 2\pi) * (1 - x_1) - sin((q_1 + Q_1 + 1/4) * 2\pi) * ((floor(Q_1 * 4)\%2) * 2 - 1) * x_2)
e^{(floor(M)+q_1)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1}
) = 0
)
))))$$

kuna Q_1'ele 2\pi liites...

$$\forall_{q_1}(\forall_{x_1}(\forall_{x_2}(\forall_{Q_1}(q_1 \in R \wedge Q_1 \in R \wedge x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \wedge \zeta(x_1 + i * x_2) = 0 \rightarrow (
\lim_{M \to \infty}(
\sum_{n=1}^{floor(e^{(floor(M)+q_1)*2\pi/x_2})}(sin(x_2 * ln(n) + Q_1 * 2\pi) * n^{-x_1})
-(sin((q_1 + Q_1) * 2\pi) * (1 - x_1) - sin((q_1 + Q_1 + 1/4) * 2\pi) * ((floor(Q_1 * 4)\%2) * 2 - 1) * x_2)
e^{(floor(M)+q_1)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1}
) = 0
\wedge \lim_{M \to \infty}(
\sum_{n=1}^{floor(e^{(floor(M)+q_1)*2\pi/x_2})}(sin(x_2 * ln(n) + Q_1 * 2\pi) * n^{-x_1})
-(sin((q_1 + Q_1) * 2\pi) * (1 - x_1) + sin((q_1 + Q_1 + 1/4) * 2\pi) * ((floor(Q_1 * 4)\%2) * 2 - 1) * x_2)
e^{(floor(M)+q_1)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1}
) = 0
)
))))$$

et see mõlemal juhul 0 oleks sin((q_1+Q_1+1/4)*2\pi)=0 seega q_1=-1/4-Q_1 või q_1=1/4-Q_1

$$\forall_{x_1}(\forall_{x_2}(\forall_{Q_1}(Q_1 \in R \wedge x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \wedge \zeta(x_1 + i * x_2) = 0 \rightarrow ($$

$$\lim_{M \to \infty}(
\sum_{n=1}^{floor(e^{(floor(M)-1/4-Q_1)*2\pi/x_2})}(sin(x_2 * ln(n) + Q_1 * 2\pi) * n^{-x_1})
+(1-x_1) * e^{(floor(M)-1/4-Q_1)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1}
) = 0$$

$$\vee \lim_{M \to \infty}(
\sum_{n=1}^{floor(e^{(floor(M)+1/4-Q_1)*2\pi/x_2})}(sin(x_2 * ln(n) + Q_1 * 2\pi) * n^{-x_1})
-(1-x_1) * e^{(floor(M)+1/4-Q_1)*2\pi*(1-x_1)/x_2} * ((1-x_1)^2 + x_2^2)^{-1}
) = 0$$

)
)))

samahästi võib m'i 2 korda aeglasemalt suurendada.

$$\forall_{x_1}(\forall_{x_2}(\forall_{Q_1}(Q_1 \in R \wedge x_1 \in R \wedge x_2 \in R \wedge x_1 > 0 \wedge \zeta(x_1 + i * x_2) = 0 \rightarrow ($$

$$\lim_{M\to\infty}($$
$$\sum_{n=1}^{floor(e^{(floor(M)/2+1/4-Q_1)*2\pi/x_2})}(sin(x_2*ln(n)+Q_1*2\pi)*n^{-x_1})$$
$$-(1-x_1)*((floor(M)\%2)*2-1)*e^{(floor(M)/2+1/4-Q_1)*2\pi*(1-x_1)/x_2}*((1-x_1)^2+x_2^2)^{-1}$$
$$)=0$$

$$)$$
$$)))$$

3 muu

3.0.1 c

$$lim_{M\to\infty}(M*(M+1)^{-x_1}-M^{1-x_1})=$$
$$x_1>=0\to return(0)$$
$$x_1<0\to return(\infty)$$

3.0.2 b

$$lim_{M\to\infty}($$
$$(sin(x_2*ln(M+1)+2*(2\pi)/4)*(1-x_1)+sin(x_2*ln(M+1)+1*(2\pi)/4)*x_2)*M*(M+1)^{-x_1}*(x_1^2-2*x_1+$$
$$x_2^2+1)^{-1}-$$
$$(sin(x_2*ln(M+1)+2*(2\pi)/4)*(1-x_1)+sin(x_2*ln(M+1)+1*(2\pi)/4)*x_2)*M^{1-x_1}*(x_1^2-2*x_1+x_2^2+1)^{-1}$$
$$)=$$
$$lim_{M\to\infty}((sin(x_2*ln(M+1)+2*(2\pi)/4)*(1-x_1)+sin(x_2*ln(M+1)+1*(2\pi)/4)*x_2)*$$
$$(M*(M+1)^{-x_1}-M^{1-x_1})$$
$$)=$$
$$x_1>0\to return(0)$$
$$x_1<=0\to return(\infty)$$

3.0.3 a

$$lim_{M\to\infty}($$
$$(sin(x_2*ln(M+1)+2*(2\pi)/4)*(1-x_1)+sin(x_2*ln(M+1)+1*(2\pi)/4)*x_2)*M*(M+1)^{-x_1}*(x_1^2-2*x_1+$$
$$x_2^2+1)^{-1}-$$
$$(sin(x_2*ln(M)+2*(2\pi)/4)*(1-x_1)+sin(x_2*ln(M)+1*(2\pi)/4)*x_2)*M^{1-x_1}*(x_1^2-2*x_1+x_2^2+1)^{-1}$$
$$)=$$
$$lim_{M\to\infty}($$
$$(sin(x_2*ln(M+1)+2*(2\pi)/4)*(1-x_1)+sin(x_2*ln(M+1)+1*(2\pi)/4)*x_2)*M*(M+1)^{-x_1}-$$
$$(sin(x_2*ln(M)+2*(2\pi)/4)*(1-x_1)+sin(x_2*ln(M)+1*(2\pi)/4)*x_2)*M^{1-x_1}$$
$$)=$$

3.0.4 d

$$lim_{M\to\infty}($$
$$(sin(x_2*ln(M+1)+2*(2\pi)/4)*(1-x_1)+sin(x_2*ln(M+1)+1*(2\pi)/4)*x_2)*M^{1-x_1}*(x_1^2-2*x_1+x_2^2+1)^{-1}-$$
$$(sin(x_2*ln(M)+2*(2\pi)/4)*(1-x_1)+sin(x_2*ln(M)+1*(2\pi)/4)*x_2)*M^{1-x_1}*(x_1^2-2*x_1+x_2^2+1)^{-1}$$
$$)=$$

$$(x_1^2-2*x_1+x_2^2+1)^{-1}*lim_{M\to\infty}(M^{1-x_1}*($$
$$(1-x_1)*(sin(x_2*ln(M+1)+2*(2\pi)/4)-sin(x_2*ln(M)+2*(2\pi)/4))+$$
$$x_2*(sin(x_2*ln(M+1)+1*(2\pi)/4)-sin(x_2*ln(M)+1*(2\pi)/4))$$
$$))=$$

$(x_1^2 - 2*x_1 + x_2^2 + 1)^{-1} * lim_{M\to\infty}(M^{1-x_1} * ($
$(cos(x_2 * ln(M+1)) - cos(x_2 * ln(M))) * x_2 -$
$(sin(x_2 * ln(M+1)) - sin(x_2 * ln(M))) * (1-x_1)))$

$lim_{M\to\infty}(M^{1-x_1} * ($
$abs((cos(x_2 * ln(M+1)) - cos(x_2 * ln(M))) * x_2)) +$
$abs((sin(x_2 * ln(M+1)) - sin(x_2 * ln(M))) * (1-x_1)))$
$))$

$lim_{M\to\infty}(M^{1-x_1} * ($
$abs((x_2 * ln(M+1) - x_2 * ln(M)) * x_2) +$
$abs((x_2 * ln(M+1) - x_2 * ln(M)) * (1-x_1))$
$))$

$lim_{M\to\infty}(M^{1-x_1} * ($
$abs((ln(M+1) - ln(M)) * x_2^2) +$
$abs((ln(M+1) - ln(M)) * (1-x_1) * x_2)$
$))$

$lim_{M\to\infty}(M^{1-x_1} * ($
$(ln(M+1) - ln(M)) * x_2^2 +$
$(ln(M+1) - ln(M)) * abs((1-x_1) * x_2)$
$))$

$lim_{M\to\infty}(M^{1-x_1} * ($
$(ln(M+1) - ln(M)) * (abs(x_2) * (abs(x_2) + abs(1-x_1)))$
$))$

$lim_{M\to\infty}(M^{1-x_1} * (ln(M+1) - ln(M)) * (abs(x_2) * (abs(x_2) + abs(1-x_1))))$

$lim_{M\to\infty}(M^{1-x_1} * (ln(M+1) - ln(M)))$

$lim_{M\to\infty}(M^{-x_1} * \ (ln(M+1) - ln(M)) * M)$

$lim_{M\to\infty}(M^{-x_1})$

$x_1 > 0 \to return\ 0$

3.0.5 f

$ln(M) * x_2 = (k * 2\pi)/2$

$M = e^{(k*2\pi)/(2*x_2)} = e^{\frac{k*2\pi}{2*x_2}}$

$k = 2 * ln(M) * x_2/(2\pi) = \frac{2*ln(M)*x_2}{2\pi}$

$x_2 = k * 2\pi/ln(M)/2 = \frac{k*2\pi}{ln(M)*2}$

asendus

$M_{uus} = e^{(floor(M)+q)*2\pi/x_2} = e^{\frac{floor(M)*2\pi}{x_2}} = e^{(floor(M)*2\pi+\alpha)/x_2}$

asendus2

$ln(M) * x_2 = (k * 2\pi) + \alpha$

M=$e^{(k*2\pi+\alpha)/(x_2)} = e^{\frac{k*2\pi+\alpha}{x_2}}$

k=(ln(M)*x$_2$ $-\alpha$)/(2π) $= \frac{ln(M)*x_2-\alpha}{2\pi}$

$x_2 = ((k*2\pi)+\alpha)/ln(M)$

q=$\alpha/2\pi$

3.0.6 g

$sin(ln(x)) =$
$\forall_x(x > -1 \wedge x < 1 \rightarrow$
$ln(x+1) = \sum_{n=1}^{\infty}((-1)^{n+1} * n^{-1} * x^n)$
)
muutuja vahetus x-1
$\forall_x(x > 0 \wedge x < 2 \rightarrow$
$ln(x) = \sum_{n=1}^{\infty}((-1)^{n+1} * n^{-1} * (x-1)^n)$
)
mv 1/x
$\forall_x(x > 1/2 \rightarrow$
$ln(1/x) = \sum_{n=1}^{\infty}((-1)^{n+1} * n^{-1} * (1/x-1)^n)$
)
mv summas
$\forall_x(x > 1/2 \rightarrow$
$ln(1/x) = \sum_{n=1}^{\infty}((-1)^{n+1} * n^{-1} * (1/x-1)^n)$
)
..

3.0.7 ln maclaurin

https://math.stackexchange.com/questions/585154/taylor-series-for-logx
valem ln_maclaurin:
$\forall_x(x > 0 \rightarrow$
$ln(x) = 2 * \sum_{k=0}^{\infty}((2*k+1)^{-1} * (x-1)^{2*k+1} * (x+1)^{-2*k-1})$
)

3.0.8 sin(ln(x)) maclaurin

sin:
$\forall_x(x \in R \rightarrow$
$sin(x) = \sum_{k=0}^{\infty}((-1)^k * x^{2*k+1} * \Gamma(2*k+2)^{-1})$
)
sin(y*ln(x)): Valem sin_y_ln_x_maclaurin_3:
$\forall_x(x > 0 \rightarrow$
$sin(y*ln(x)) = \sum_{k_1=0}^{\infty}((\sum_{k_2=0}^{\infty}((2*k_2+1)^{-1} * (x-1)^{2*k_2+1} * (x+1)^{-2*k_2-1}))^{2*k_1+1} * y^{2*k_1+1} * factorial(2*k_1+1)^{-1} *$
$2^{2*k_1+1} * (-1)^{k_1})$
)
kasutan Multinomial theoreemi. Siin on eeldatud, et True'ga korrutamine on nagu 1ega korrutamine ja Falsega korrutamine on nagu 0'iga korrutamine. Valem sin_y_ln_x_multinomial_1:
$\forall_x(x > 0 \rightarrow$
$sin(y*ln(x)) = lim_{M\to\infty}(\sum_{k_1=0}^{M-1}(\sum_{k_2=0}^{(2*k_1+2)^M-1}($
$(\sum_{k_3=0}^{M-1}(\ floor(k_2/(2*k_1+2)^{k_3})\%(2*k_1+2)\ ) = 2*k_1+1) *$
$\Pi_{k_3=0}^{M-1}(factorial(\ floor(k_2/(2*k_1+2)^{k_3})\%(2*k_1+2)\ )^{-1} * ((2*k_3+1)^{-1} * (x-1)^{2*k_3+1} * (x+1)^{-2*k_3-1})^{floor(k_2/(2*k_1+2)^{k_3})\%(2*k_1+2)}$
$) * y^{2*k_1+1} * 2^{2*k_1+1} * (-1)^{k_1}))$
)

elementide korrutamise asemel võib nende astemed liita:

$$\forall_x(x>0\rightarrow sin(y*ln(x))=$$
$$lim_{M\rightarrow\infty}(\sum_{k_1=0}^{M-1}(\sum_{k_2=0}^{(2*k_1+2)^M-1}($$
$$(\sum_{k_3=0}^{M-1}(\ floor(k_2/(2*k_1+2)^{k_3})\%(2*k_1+2)\ )=2*k_1+1)*$$
$$\Pi_{k_3=0}^{M-1}(factorial(floor(k_2/(2*k_1+2)^{k_3})\%(2*k_1+2))^{-1}*(2*k_3+1)^{-(floor(k_2/(2*k_1+2)^{k_3})\%(2*k_1+2))})*((x-1)/(x+$$
$$1))^{\sum_{k_3=0}^{M-1}((2*k_3+1)*(floor(k_2/(2*k_1+2)^{k_3})\%(2*k_1+2)))}*y^{2*k_1+1}*2^{2*k_1+1}*(-1)^{k_1})))$$
$$)$$

lihtsustan. Valem sin_y_ln_x_multinomial:

$$\forall_x(x>0\rightarrow sin(y*ln(x))=$$
$$lim_{M\rightarrow\infty}(\sum_{k_1=0}^{M-1}(y^{2*k_1+1}*2^{2*k_1+1}*(-1)^{k_1}*\sum_{k_2=0}^{(2*k_1+2)^M-1}($$
$$(\sum_{k_3=0}^{M-1}(\ floor(k_2/(2*k_1+2)^{k_3})\%(2*k_1+2)\ )=2*k_1+1)*$$
$$(x-1)^{\sum_{k_3=0}^{M-1}((2*k_3+1)*(floor(k_2/(2*k_1+2)^{k_3})\%(2*k_1+2)))}*$$
$$(x+1)^{-\sum_{k_3=0}^{M-1}((2*k_3+1)*(floor(k_2/(2*k_1+2)^{k_3})\%(2*k_1+2)))}*$$

$$\Pi_{k_3=0}^{M-1}(factorial(floor(k_2/(2*k_1+2)^{k_3})\%(2*k_1+2))^{-1}*(2*k_3+1)^{-(floor(k_2/(2*k_1+2)^{k_3})\%(2*k_1+2))})$$
$$)))$$
$$)$$

viin teguri sulgude sisse:

$$\forall_x(x>0\rightarrow sin(y*ln(x))=$$
$$lim_{M\rightarrow\infty}(\sum_{k_1=0}^{M-1}(\sum_{k_2=0}^{(2*k_1+2)^M-1}($$
$$(\sum_{k_3=0}^{M-1}(\ floor(k_2/(2*k_1+2)^{k_3})\%(2*k_1+2)\ )=2*k_1+1)*$$
$$y^{2*k_1+1}*2^{2*k_1+1}*(-1)^{k_1}*$$
$$(x-1)^{\sum_{k_3=0}^{M-1}((2*k_3+1)*(floor(k_2/(2*k_1+2)^{k_3})\%(2*k_1+2)))}*$$
$$(x+1)^{-\sum_{k_3=0}^{M-1}((2*k_3+1)*(floor(k_2/(2*k_1+2)^{k_3})\%(2*k_1+2)))}*$$
$$\Pi_{k_3=0}^{M-1}(factorial(\ floor(k_2/(2*k_1+2)^{k_3})\%(2*k_1+2)\ )^{-1}*(2*k_3+1)^{-(floor(k_2/(2*k_1+2)^{k_3})\%(2*k_1+2))})$$
$$)))$$
$$)$$

selle asemel et üle eri n2'e ristsummade summeeridaarvutan lihtsalt n2'e ristsumma. Valem sin_y_ln_x_multinomial_2:

SIIN VIGA!

(2*k_2+1)^{-(floor(k_1/(2*M)^{k_2})\%(2*M))}

$$\forall_x(x>0\rightarrow sin(y*ln(x))=$$
$$lim_{M\rightarrow\infty}(\sum_{k_1=0}^{(2*M)^M-1}($$
$$(\sum_{k_2=0}^{M-1}(floor(k_1/(2*M)^{k_2})\%(2*M))\%2)*$$
$$(\sum_{k_2=0}^{M-1}(floor(k_1/(2*M)^{k_2})\%(2*M))<2*M)*$$
$$(y*2)^{\sum_{k_2=0}^{M-1}(floor(k_1/(2*M)^{k_2})\%(2*M))}*$$
$$(-1)^{(\sum_{k_2=0}^{M-1}(floor(k_1/(2*M)^{k_2})\%(2*M))-1)/2}*$$
$$(x-1)^{\sum_{k_2=0}^{M-1}((2*k_2+1)*(floor(k_1/(2*M)^{k_2})\%(2*M)))}*$$
$$(x+1)^{-\sum_{k_2=0}^{M-1}((2*k_2+1)*(floor(k_1/(2*M)^{k_2})\%(2*M)))}*$$
$$\Pi_{k_2=0}^{M-1}(factorial(floor(k_1/(2*M)^{k_2})\%(2*M))^{-1}*(2*k_2+1)^{-(floor(k_1/(2*M)^{k_2})\%(2*M))})$$
$$))$$
$$)$$

kuna Mi lähenedes lõpmatusele see avaldis nagu nii koondub ja selle muutuse tegemine ainult lisab liikmeid, mis on absoluutväärtuselt väiksemad kui oleksid liikmed, mis lisanduksid Mi suurendamisel. Valem sin_y_ln_x_multinomial_4:

$$\forall_x(x>0\rightarrow sin(y*ln(x))=$$
$$lim_{M\rightarrow\infty}(\sum_{k_1=0}^{(2*M)^M-1}($$
$$(\sum_{k_2=0}^{M-1}(floor(k_1/(2*M)^{k_2})\%(2*M))\%2)*$$

$(y*2)^{\sum_{k_2=0}^{M-1}(floor(k_1/(2*M)^{k_2})\%(2*M))}*$
$(-1)^{(\sum_{k_2=0}^{M-1}(floor(k_1/(2*M)^{k_2})\%(2*M))+3)/2}*$
$(x-1)^{\sum_{k_2=0}^{M-1}((2*k_2+1)*(floor(k_1/(2*M)^{k_2})\%(2*M)))}*$
$(x+1)^{-\sum_{k_2=0}^{M-1}((2*k_2+1)*(floor(k_1/(2*M)^{k_2})\%(2*M)))}*$
$\Pi_{k_2=0}^{M-1}(factorial(floor(k_1/(2*M)^{k_2})\%(2*M))^{-1}*(2*k_2+1)^{-(floor(k_1/(2*M)^{k_2})\%(2*M))})$
))
)

kasutan multimomial astendajategenereerimiseks teist valemit $v[A]=\sum_{k_4=0}^{\infty}(k_2/2^{k_4*M+A}\%2)$. Ehk biti indexi jääk määrab, et mitmenda astendaja bit see on.

3.0.9 cos(ln(x)) maclaurin

cos:

$\forall_x(x\in R\rightarrow$
$cos(x)=\sum_{k=0}^{\infty}((-1)^k*x^{2*k}*factorial(2*k)^{-1})$
)

cos(y*ln(x)):

$\forall_x(x>0\rightarrow$
$cos(y*ln(x))=\sum_{k_1=0}^{\infty}((-1)^{k_1}*(2*\sum_{k_2=0}^{\infty}((2*k_2+1)^{-1}*(x-1)^{2*k_2+1}*(x+1)^{-2*k_2-1}))^{2*k_1}*y^{2*k_1}*\Gamma(2*k_1+1)^{-1})$
)

Valem cos_y_ln_x_multinomial:

$\forall_x(x>0\rightarrow cos(y*ln(x))=$
$lim_{M\rightarrow\infty}(\sum_{k_1=0}^{M-1}(y^{2*k_1}*2^{2*k_1}*(-1)^{k_1}*\sum_{k_2=0}^{(2*k_1+1)^M-1}($
$(\sum_{k_3=0}^{M-1}(\ floor(k_2/(2*k_1+1)^{k_3})\%(2*k_1+1)\ )=2*k_1)*$
$(x-1)^{\sum_{k_3=0}^{M-1}((2*k_3+1)*(floor(k_2/(2*k_1+1)^{k_3})\%(2*k_1+1)))}*$
$(x+1)^{-\sum_{k_3=0}^{M-1}((2*k_3+1)*(floor(k_2/(2*k_1+1)^{k_3})\%(2*k_1+1)))}*$
$\Pi_{k_3=0}^{M-1}(factorial(\ floor(k_2/(2*k_1+1)^{k_3})\%(2*k_1+1)\ )^{-1}*(2*k_3+1)^{-floor(k_2/(2*k_1+1)^{k_3})\%(2*k_1+1)})$
)))
)

Valem cos_y_ln_x_multinomial_2:

$\forall_x(x>0\rightarrow cos(y*ln(x))=$
$lim_{M\rightarrow\infty}(\sum_{k_1=0}^{(2*M-1)^M-1}($
$((1+\sum_{k_2=0}^{M-1}(floor(k_1/(2*M-1)^{k_2})\%(2*M-1)))\%2)*$
$(\sum_{k_2=0}^{M-1}(floor(k_1/(2*M-1)^{k_2})\%(2*M-1))<2*M-1)*$
$(y*2)^{\sum_{k_2=0}^{M-1}(floor(k_1/(2*M-1)^{k_2})\%(2*M-1))}*$
$(-1)^{\sum_{k_2=0}^{M-1}(floor(k_1/(2*M-1)^{k_2})\%(2*M-1))/2}*$
$(x-1)^{\sum_{k_2=0}^{M-1}((2*k_2+1)*(floor(k_1/(2*M-1)^{k_2})\%(2*M-1)))}*$
$(x+1)^{-\sum_{k_2=0}^{M-1}((2*k_2+1)*(floor(k_1/(2*M-1)^{k_2})\%(2*M-1)))}*$
$\Pi_{k_2=0}^{M-1}(factorial(floor(k_1/(2*M-1)^{k_2})\%(2*M-1))^{-1}*(2*k_2+1)^{-(floor(k_1/(2*M-1)^{k_2})\%(2*M-1))})$
))
)

positsioonsüsteemi, millega eksponendid on kodeeritud on nüüd 2*M-1 asemel 2*M. veel üks valem:

$\forall_x(x>0\rightarrow cos(y*ln(x))=$
$lim_{M\rightarrow\infty}(\sum_{n_1=0}^{(2*M)^M-1}($
$((1+\sum_{n_2=0}^{M-1}(floor(n_1/(2*M)^{n_2})\%(2*M)))\%2)*$
$(\sum_{n_2=0}^{M-1}(floor(n_1/(2*M)^{n_2})\%(2*M))<2*M-1)*$
$(y*2)^{\sum_{n_2=0}^{M-1}(floor(n_1/(2*M)^{n_2})\%(2*M))}*$

$(-1)^{\sum_{n_2=0}^{M-1}(floor(n_1/(2*M)^{n_2})\%(2*M))/2} *$
$(x-1)^{\sum_{n_2=0}^{M-1}((2*n_2+1)*(floor(n_1/(2*M)^{n_2})\%(2*M)))} *$
$(x+1)^{-\sum_{n_2=0}^{M-1}((2*n_2+1)*(floor(n_1/(2*M)^{n_2})\%(2*M)))} *$
$\Pi_{n_2=0}^{M-1}(factorial(floor(n_1/(2*M)^{n_2})\%(2*M))^{-1} * (2*n_2+1)^{-(floor(n_1/(2*M)^{n_2})\%(2*M))})$
))
)

Valem cos_y_ln_x_multinomial_4_2:

$\forall_x(x>0 \rightarrow cos(y*ln(x)) =$
$lim_{M\rightarrow\infty}(\sum_{n_1=0}^{(2*M)^M-1}($
$((1+\sum_{n_2=0}^{M-1}(floor(n_1/(2*M)^{n_2})\%(2*M)))\%2) * (y*2)^{\sum_{n_2=0}^{M-1}(floor(n_1/(2*M)^{n_2})\%(2*M))} *$
$(-1)^{\sum_{n_2=0}^{M-1}(floor(n_1/(2*M)^{n_2})\%(2*M))/2} *$
$(x-1)^{\sum_{n_2=0}^{M-1}((2*n_2+1)*(floor(n_1/(2*M)^{n_2})\%(2*M)))} *$
$(x+1)^{-\sum_{n_2=0}^{M-1}((2*n_2+1)*(floor(n_1/(2*M)^{n_2})\%(2*M)))} *$
$\Pi_{n_2=0}^{M-1}(factorial(floor(n_1/(2*M)^{n_2})\%(2*M))^{-1} * (2*n_2+1)^{-(floor(n_1/(2*M)^{n_2})\%(2*M))})$
))
)

3.0.10 sisemine summa

$A = \sum_{n_2=0}^{M-1}((2*n_2+1)*(floor(n_0/(2*M)^{n_2})\%(2*M)))$

$-$
$\sum_{n_1=0}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})-1}($
$n_1^A *$
$(n_1+1)^{-x_1} *$
$(n_1+2)^{-A} *$
)
kasutan binomial valemit:$\forall_n(\forall_a(\forall_b((a+b)^n = \sum_{k=0}^{\infty}(faktorial(n)*faktorial(k)^{-1}*faktorial(n-k)^{-1}*a^k*b^{n-k})$

saadud tulemus:

$\sum_{n_1=0}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})-1}($
$\sum_{k=0}^{\infty}(faktorial(-x_1)*faktorial(k)^{-1}*faktorial(-x_1-k)^{-1}*n_1^{k+A}) *$
$\sum_{k=0}^{\infty}(faktorial(-A)*faktorial(k)^{-1}*faktorial(-A-k)^{-1}*n_1^k*2^{-A-k}) *$
)

panen 1 summa 2 sisse:

$\sum_{n_1=0}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})-1}($
$\sum_{k_1=0}^{\infty}(\sum_{k_2=0}^{\infty}(faktorial(-x_1)*faktorial(k_1)^{-1}*faktorial(-x_1-k_1)^{-1}*n_1^{k_1+A}*faktorial(-A)*faktorial(k_2)^{-1}*$
$faktorial(-A-k_2)^{-1}*n_1^{k_2}*2^{-A-k_2})) *$
)

lihtsustan seest ja viin n1 summa sisse:

$\sum_{k_1=0}^{\infty}(\sum_{k_2=0}^{\infty}(faktorial(-x_1)*faktorial(k_1)^{-1}*faktorial(-x_1-k_1)^{-1}*faktorial(-A)*faktorial(k_2)^{-1}*faktorial(-A-k_2)^{-1}*2^{-A-k_2}*$
$\sum_{n_1=0}^{floor(e^{(floor(M)/abs(x_2)+q_2/x_2)*2\pi})-1}(n_1^{k_1+k_2+A})$
))

3.0.11 veel

$\int_a^b (dx * sin(ln(x))) = (-1/2 * b * (cos(log(b)) - sin(log(b)))) - (-1/2 * a * (cos(log(a)) - sin(log(a)))) = (b/2 * (sin(ln(b)) - cos(ln(b)))) - (a/2 * (sin(ln(a)) - cos(ln(a)))) = 2^{-1/2} * (b * sin(ln(b) - 2\pi/8) - a * sin(ln(a) - 2\pi/8)) = 2^{-1/2} * (a * cos(ln(a)) - b * cos(ln(b)))$

veel

$\int_a^{a+b} (dx * sin(ln(x))) = 2^{-1/2} * (a * cos(ln(a)) - (a+b) * cos(ln(a+b))) = 2^{-1/2} * (a * (cos(ln(a)) - cos(ln(a+b))) - b * cos(ln(a+b)))$

veel

$\int_{floor(e^{(floor(M)/2+1/4)*2\pi/abs(x_2)-arctan((1-x_1)/x_2)/x_2})}^{floor(e^{(floor(M)/2+1/4)*2\pi/abs(x_2)-arctan((1-x_1)/x_2)/x_2})+b} (dx * sin(ln(x))) = 2^{-1/2} * (floor(e^{(floor(M)/2+1/4)*2\pi/abs(x_2)-arctan((1-x_1)/x_2)/x_2}) * (cos(ln(floor(e^{(floor(M)/2+1/4)*2\pi/abs(x_2)-arctan((1-x_1)/x_2)/x_2}))) - cos(ln(floor(e^{(floor(M)/2+1/4)*2\pi/abs(x_2)-arctan((1-x_1)/x_2)/x_2}) + b))) - b * cos(ln(floor(e^{(floor(M)/2+1/4)*2\pi/abs(x_2)-arctan((1-x_1)/x_2)/x_2}) + b))) = 2^{-1/2} * (floor(e^{(floor(M)/2+1/4)*2\pi/abs(x_2)-arctan((1-x_1)/x_2)/x_2}) * (cos((floor(M)/2 + 1/4) * 2\pi/abs(x_2) - arctan((1 - x_1)/x_2)/x_2) - cos(ln(floor(e^{(floor(M)/2+1/4)*2\pi/abs(x_2)-arctan((1-x_1)/x_2)/x_2}) + b))) - b * cos(ln(floor(e^{(floor(M)/2+1/4)*2\pi/abs(x_2)-arctan((1-x_1)/x_2)/x_2}) + b)))$

3.0.12 integral

siinuse oma $\int_A^B (dy * sin(x_2 * ln(y) + (2\pi) * 2/4) * y^{-x_1}) =$

$-\int_A^B (dy * sin(x_2 * ln(y)) * y^{-x_1}) =$

$\frac{B^{1-x_1} * ((x_1-1) * sin(x_2 * ln(B)) + x_2 * cos(x_2 * ln(B)))}{x_1^2 - 2 * x_1 + x_2^2 + 1} - \frac{A^{1-x_1} * ((x_1-1) * sin(x_2 * ln(A)) + x_2 * cos(x_2 * ln(A)))}{x_1^2 - 2 * x_1 + x_2^2 + 1} =$

$(B^{1-x_1} * ((x_1 - 1) * sin(x_2 * ln(B)) + x_2 * cos(x_2 * ln(B))) - A^{1-x_1} * ((x_1 - 1) * sin(x_2 * ln(A)) + x_2 * cos(x_2 * ln(A)))) * ((1 - x_1)^2 + x_2^2)^{-1} =$

$(B^{1-x_1} * (x_1 - 1) * sin(x_2 * ln(B)) + B^{1-x_1} * x_2 * cos(x_2 * ln(B)) - A^{1-x_1} * (x_1 - 1) * sin(x_2 * ln(A)) - A^{1-x_1} * x_2 * cos(x_2 * ln(A))) * ((1 - x_1)^2 + x_2^2)^{-1} =$

$(-B^{1-x_1} * (1 - x_1) * sin(x_2 * ln(B)) + B^{1-x_1} * x_2 * cos(x_2 * ln(B)) + A^{1-x_1} * (1 - x_1) * sin(x_2 * ln(A)) - A^{1-x_1} * x_2 * cos(x_2 * ln(A))) * ((1 - x_1)^2 + x_2^2)^{-1} =$

$(B^{1-x_1} * (-(1 - x_1) * sin(x_2 * ln(B)) + x_2 * cos(x_2 * ln(B))) + A^{1-x_1} * ((1 - x_1) * sin(x_2 * ln(A)) - x_2 * cos(x_2 * ln(A)))) * ((1 - x_1)^2 + x_2^2)^{-1} =$

$(B^{1-x_1} * \sqrt{((1 - x_1)^2 + x_2^2)} * sin(x_2 * ln(B) - arctan(x_2/(1 - x_1))) + A^{1-x_1} * \sqrt{((1 - x_1)^2 + x_2^2)} * sin(x_2 * ln(A) - arctan(x_2/(1 - x_1)))) * ((1 - x_1)^2 + x_2^2)^{-1} =$

$(B^{1-x_1} * sin(x_2 * ln(B) - arctan(x_2/(1 - x_1))) + A^{1-x_1} * sin(x_2 * ln(A) - arctan(x_2/(1 - x_1)))) * ((1 - x_1)^2 + x_2^2)^{-1/2}$

coosinuse oma $\int_A^B (dy * sin(x_2 * ln(y) + (2\pi) * 1/4) * y^{-x_1}) =$

$\int_A^B (dy * cos(x_2 * ln(y)) * y^{-x_1}) =$

$\frac{B^{1-x_1} * (x_2 * sin(x_2 * ln(B)) - (x_1 - 1) * cos(x_2 * ln(B)))}{x_1^2 - 2 * x_1 + x_2^2 + 1} - \frac{A^{1-x_1} * (x_2 * sin(x_2 * ln(A)) - (x_1 - 1) * cos(x_2 * ln(A)))}{x_1^2 - 2 * x_1 + x_2^2 + 1} =$

$(B^{1-x_1} * (x_2 * sin(x_2 * ln(B)) - (x_1 - 1) * cos(x_2 * ln(B))) - A^{1-x_1} * (x_2 * sin(x_2 * ln(A)) - (x_1 - 1) * cos(x_2 * ln(A)))) * ((1 - x_1)^2 + x_2^2)^{-1} =$

$(B^{1-x_1} * x_2 * sin(x_2 * ln(B)) - B^{1-x_1} * (x_1 - 1) * cos(x_2 * ln(B)) - A^{1-x_1} * x_2 * sin(x_2 * ln(A)) + A^{1-x_1} * (x_1 - 1) * cos(x_2 * ln(A))) * ((1 - x_1)^2 + x_2^2)^{-1} =$

$(B^{1-x_1} * x_2 * sin(x_2 * ln(B)) + B^{1-x_1} * (1 - x_1) * cos(x_2 * ln(B)) - A^{1-x_1} * x_2 * sin(x_2 * ln(A)) - A^{1-x_1} * (1 - x_1) * cos(x_2 * ln(A))) * ((1 - x_1)^2 + x_2^2)^{-1} =$

$(B^{1-x_1} * (x_2 * sin(x_2 * ln(B)) + (1 - x_1) * cos(x_2 * ln(B))) - A^{1-x_1} * (x_2 * sin(x_2 * ln(A)) - (1 - x_1) * cos(x_2 * ln(A)))) * ((1 - x_1)^2 + x_2^2)^{-1} =$

$(B^{1-x_1} * sin(x_2 * ln(B) + arctan((1 - x_1)/x_2)) - A^{1-x_1} * sin(x_2 * ln(A) + arctan((1 - x_1)/x_2))) * ((1 - x_1)^2 + x_2^2)^{-1/2}$